Off on a Tangent

Meditations on the derivative

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Introduction

Prerequisites

This text assumes basic familiarity with point-set topology (specifically, metric spaces) and with linear algebra. At a few points, there may be some ideas from point-set topology and linear algebra that you have not encountered before. A sort of "bare minimum" exposition of some of these ideas is included in chapter 0.

How to use this book

You are advised to *not* spend time reading chapter 0 thoroughly before jumping into the main part of the text (chapter 1 onwards). When ideas from chapter 0 are invoked in the main part, a reference to the relevant part of chapter 0 is included. My suggestion is to only look at chapter 0 when you run into a reference to it. That said, when you are looking at a part of chapter 0, make sure you solve try to all the relevant exercises that appear there.

Suggestions for improvement

This is a first draft, and there are bound to be many errors. Please be on the lookout for them! If you think you've found one, please share it with me.

I'll also be very interested to hear about non-errors: sentences that are maybe correct but phrased confusingly, places where adding a picture could be helpful, etc.

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0. Prerequisites

0.A. Little-oh notation

Throughout this section, let X be a metric space and let x_0 be a point in X.

Definition 0.1. A function $g : X \setminus \{x_0\} \to \mathbb{R}$ is *positive* if $g(x) > 0$ for all $x \in X \setminus \{x_0\}$. Similarly, g is *non-negative* if $g(x) \geqslant 0$ for all $x \in X \setminus \{x_0\}$.

Suppose g is a positive function. Definition 0.2 below formulates what it means for a non-negative function $f : X \setminus \{x_0\} \to \mathbb{R}$ to be "little-oh of g as x tends to x_0." Intuitively, this condition means that f is *much smaller* than g near x_0.

Definition 0.2. A function $f : X \setminus \{x_0\} \to \mathbb{R}$ is *little-oh of* g *as* x *tends to* x_0, written

$$f(x) = o(g(x)) \text{ as } x \to x_0$$

if, for every $\epsilon > 0$, there exists an open neighborhood U of x_0 such that $f(x) \leqslant \epsilon g(x)$ for all $x \in U \setminus \{x_0\}$. Equivalently, this means that

$$\lim_{x \to x_0} \frac{|f(x)|}{g(x)} = 0.$$

When x_0 can be inferred from context, we write simply $f = o(g)$.

Remark 0.3. It's worth noting that the use of the symbol "=" above is mathematically abusive. The left-hand side of the "=" is a function and the right-hand side is a property of functions; of course, a function cannot literally be equal to a property. Rather, the "=" is being used here to mean that the function on the left-hand side *has* the property described on the right-hand side. As annoying as it is, this abuse of notation is fairly standard, so it's probably best to just get used to it.

Exercise 0.4. Prove that the two conditions in definition 0.2 are in fact equivalent.

Exercise 0.5. Let $X = \mathbb{R}$ and $x_0 = 0$. For each of the functions f and g described below, sketch graphs of f and g and then determine whether or not $f = o(g)$ as $x \to 0$.

(a) $f(x) = x$ and $g(x) = x^2$.

(b) $f(x) = x$ and $g(x) = |x|$.

(c) $f(x) = x^2$ and $g(x) = |x|$.

Exercise 0.6. Let $X = \mathbb{R}^2$ and $x_0 = 0$. For each of the following functions f, determine whether or not $f(x) = o(|x|)$ as $x \to 0$.

(a) $f(x, y) = x - y$.

(b) $f(x, y) = x^2 + y^2$.

Exercise 0.7 ("Smaller than small is still small")**.** Suppose $f_1, f_2 : X \setminus \{x_0\} \to \mathbb{R}$ are functions such that $|f_1(x)| \leqslant |f_2(x)|$ for all x in a punctured neighborhood of x_0, and $f_2(x) = o(g(x))$ as $x \to x_0$. Then $f_1(x) = o(g(x))$ also.

Exercise 0.8. For a fixed positive function $g : X \setminus \{x_0\} \to \mathbb{R}$, prove that the set of functions $f : X \setminus \{x_0\} \to \mathbb{R}$ which are $o(g)$ as $x \to x_0$ is a vector space under the natural operations. In other words, verify the following three facts.

(1) The zero function is $o(g(x))$.

(2) ("Scalar multiples of small are still small") If $c \in \mathbb{R}$ is a scalar and $f(x) = o(g(x))$, then

$$cf(x) = o(g(x)).$$

(3) ("Sum of smalls is still small") If $f_1(x) = o(g(x))$ and $f_2(x) = o(g(x))$, then

$$(f_1 + f_2)(x) = o(g(x)).$$

Exercise 0.9. Suppose $f_1, f_2, g_1, g_2 : X \setminus \{x_0\} \to \mathbb{R}$ are functions with g_1, g_2 positive such that $f_1(x) = o(g_1(x))$ and $f_2(x) = o(g_2(x))$ as $x \to x_0$. Prove that $(f_1 \cdot f_2)(x) = o((g_1 \cdot g_2)(x))$.

0.B. Linear algebra

0.B.1. Important vector spaces

Definition 0.10. We write $\mathbb{R}^n$ to denote the vector space of lists of n numbers. We will sometimes write its elements as horizontal lists with commas, as in $(a_1, \ldots, a_n)$, and

sometimes as vertical column vectors, as in

$$\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}.$$

Definition 0.11. We write $\mathcal{M}_{n,m}$ for the set of all $n \times m$ matrices (ie, the matrices with n rows and m columns). This is naturally a vector space.

Definition 0.12. If V and W are both vector spaces, we write $\mathcal{L}(V, W)$ for the set of all linear maps $V \to W$. This is naturally a vector space.

0.B.2. Matrices

Definition 0.13. Suppose V is a finite dimensional vector space and $B = (v_1, \ldots, v_n)$ is a basis for V. Then any $v \in V$ can be written uniquely as $a_1 v_1 + \cdots + a_n v_n$ for some scalars $a_1, \ldots, a_n \in \mathbb{R}$, and we define the *representation of v with respect to* B, denoted $[v]_B$, to be the column vector with n entries that records the scalars $a_1, \ldots, a_n$. In other words,

$$[v]_B = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}.$$

Lemma 0.14. *If V is an n-dimensional vector space and $B = (v_1, \ldots, v_n)$ is a basis for V, then the function $V \to \mathbb{R}^n$ given by $v \mapsto [v]_B$ is an isomorphism, with inverse given by*

$$\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \mapsto a_1 v_1 + \cdots + a_n v_n.$$

Definition 0.15. Suppose $\ell : V \to W$ is a linear map between finite dimensional vector spaces. Let $B = v_1, \ldots, v_m$ be a basis for V and let C be a basis for W. The *matrix representation of ℓ with respect to* B *and* C, denoted $[\ell]_{B,C}$, is the $n \times m$ matrix whose ith column of $[\ell]_{B,C}$ is $[\ell(v_i)]_C$. In other words,

$$[\ell]_{B,C} = \begin{bmatrix} [\ell(v_1)]_C & \cdots & [\ell(v_m)]_C \end{bmatrix}.$$

Lemma 0.16. *If $\ell : V \to W$ is a linear map between finite dimensional vector spaces and B and C are bases for V and W, respectively, then*

$$[\ell]_{B,C}[v]_B = [\ell(v)]_C$$

for any $v \in V$.

Lemma 0.17. *If* $\ell : U \to V$ *and* $\ell' : V \to W$ *are linear maps between finite dimensional vector spaces, and* A, B *and* C *are bases for* U, V*, and* W*, respectively, then*

$$[\ell' \circ \ell]_{A,C} = [\ell']_{B,C}[\ell]_{A,B}.$$

Lemma 0.18. *If* $\ell : V \to W$ *is a linear map between finite dimensional vector spaces and* B *and* C *are bases for* V *and* W*, respectively, then the rank of the matrix* $[\ell]_{B,C}$ *equals the dimension of the range* $\ell(V)$.

Lemma 0.19. *Suppose that* V *and* W *are* m*- and* n*-dimensional vector spaces, respectively, and that and* B *and* C *are bases for* V *and* W*, respectively. Then the function* $\ell \mapsto [\ell]_{B,C}$ *is an isomorphism*

$$\mathcal{L}(V, W) \longrightarrow \mathcal{M}_{n,m}.$$

Definition 0.20. If $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map, the *standard matrix representation* of ℓ, denoted $[\ell]$, is the matrix of ℓ with respect to the standard bases on $\mathbb{R}^m$ and $\mathbb{R}^n$.

Lemma 0.21. *If* A *is a* $n \times m$ *matrix and* $\ell_A : \mathbb{R}^m \to \mathbb{R}^n$ *is the linear map* $\ell_A(v) = Av$*, then*

$$[\ell_A] = A.$$

0.C. Norms of linear maps

Definition 0.22. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. Then we define the *operator norm*, or just *norm*, of ℓ, denoted $\|\ell\|$, by

$$\|\ell\| = \sup\{|\ell(v)| : |v| \leqslant 1\}.$$

Exercise 0.23. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. Show that

$$\|\ell\| = \inf\{\lambda \in \mathbb{R} : |\ell(v)| \leqslant \lambda|v| \text{ for all } v \in \mathbb{R}^m\}.$$

In particular, $|\ell(v)| \leqslant \|\ell\| \cdot |v|$ for all $v \in \mathbb{R}^m$.

Exercise 0.24. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. Show that $\ell = 0$ if and only if $\|\ell\| = 0$.

Exercise 0.25 (Subadditivity)**.** Suppose $\ell, \ell' : \mathbb{R}^m \to \mathbb{R}^n$ are both linear maps. Show that

$$\|\ell + \ell'\| \leqslant \|\ell\| + \|\ell'\|.$$

Definition 0.26. Let $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ denote the set of all linear maps $\mathbb{R}^m \to \mathbb{R}^n$. Define the distance between $\ell, \ell' \in \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ by $\|\ell - \ell'\|$. It follows from exercises 0.24 and 0.25 that this distance function makes $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ a metric space.

Exercise 0.27 ("Evaluation is continuous")**.** Suppose $v_0 \in \mathbb{R}^m$ is a fixed vector. Then the "evaluate at v_0" function $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \to \mathbb{R}^n$ given by $\phi \mapsto \phi(v_0)$ is continuous. *Possible hint*. Since the "evaluate at v_0" function is linear, it is sufficient to check continuity at 0.

Exercise 0.28 (Submultiplicativity)**.** Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ and $\ell' : \mathbb{R}^n \to \mathbb{R}^p$ are both linear maps. Show that

$$\|\ell' \circ \ell\| \leqslant \|\ell'\| \cdot \|\ell\|.$$

Give an example to show that this inequality can be strict.

Exercise 0.29 ("Composition is continuous")**.** (a) Suppose $\ell : \mathbb{R}^n \to \mathbb{R}^p$ is a linear map. Show that the "postcompose with ℓ" function $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^p)$ given by $\phi \mapsto \ell \circ \phi$ is continuous.

(b) Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. Show that the "precompose with ℓ" function $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^p) \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^p)$ given by $\phi \mapsto \phi \circ \ell$ is continuous.

Possible hint (for both parts). Because both the "postcompose with ℓ" and the "precompose with ℓ" maps are linear, it is sufficient to show that the maps are continuous at 0. You will likely find exercise 0.28 useful.

Exercise 0.30 ("Linear functions of small functions are still small")**.** Suppose $r : \mathbb{R}^m \to \mathbb{R}^n$ is a function such that $|r(v)| = o(|v|)$ as $v \to 0$. If $\ell : \mathbb{R}^n \to \mathbb{R}^p$ is linear, prove that $|\ell(r(v))| = o(|v|)$ also.

Lemma 0.31. *Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a nonzero linear map. Then there exists $v_0 \in \mathbb{R}^m$ such that $|v_0| = 1$ and $|\ell(v_0)| = \|\ell\|$.*

Proof. The "unit disk"

$$D = \{x \in \mathbb{R}^m : |x| \leqslant 1\}$$

is a nonempty, connected, and compact subset of $\mathbb{R}^m$. The function $v \mapsto |\ell(v)|$ is also a continuous function $D \to \mathbb{R}$, so its image $|\ell(D)|$ must also be a nonempty, connected, and compact subset of $\mathbb{R}$. In other words, it must be a closed interval in $\mathbb{R}$. This means that there exists $v_0 \in D$ such that

$$|\ell(v_0)| = \sup |\ell(D)| = \|\ell\|.$$

Since $\ell \neq 0$, we know $\|\ell\| \neq 0$ from exercise 0.24, which in turn means that $v_0 \neq 0$. To prove that $|v_0| = 1$, assume for a contradiction that $|v_0| < 1$, and let $c = 1/|v_0| > 1$. Then $|cv_0| = |c||v_0| = 1$, so $cv_0 \in D$. But then

$$|\ell(cv_0)| = |c\ell(v_0)| = c|\ell(v_0)| > |\ell(v_0)| = \sup |\ell(D)|,$$

which is absurd. □

Exercise 0.32. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map and $|\ell(v)| = o(|v|)$ as $v \to 0$. Prove that $\ell = 0$.

0.D. Topological spaces

This is a bare-bones introduction to topological spaces, intended for readers who have seen metric spaces but not topological spaces before.

0.D.1. Basics

Definition 0.33. Let X be a set. A *topology* on X is a set τ_X of subsets of X, called *open subsets*, which satisfy the following axioms.

(T1) $\emptyset$ and X are both open subsets.

(T2) The union of an arbitrary collection of open subsets is open.

(T3) The intersection of a finite collection of open subsets is open.

A pair (X, τ_X) consisting of a set X and a topology τ on X is called a *topological space*. Often, we write simply "X" in place of the pair (X, τ_X).

Topological spaces come up throughout mathematics, and in general can be very very bizarre. We will not delve into the general theory of topological spaces, and won't see any of these bizarre examples. Instead, you are encouraged to content yourself with the following examples.

Example 0.34 (Metric spaces)**.** If X is a metric space, the collection of subsets of X that are open with respect to the metric defines a topology on X. Thus, every metric space can be regarded as a topological space in a natural way. It's worth noticing that this process of regarding a metric space as a topological space "forgets information," since equivalent metrics on a set will define the same topology.

Example 0.35 (Quotient spaces)**.** Suppose Y is any topological space (for instance, one that comes from a metric space), and let $\sim$ be an equivalence relation on Y. For an element $y \in Y$, let $[y]$ denote its equivalence class, and let $X = Y/\sim$ denote the set of all equivalence classes. There is a natural function $\pi : Y \to X$ given by sending a point $y \in Y$ to its equivalence class $[y]$. We define a topology on X, called the *quotient topology*, by declaring a subset $U \subseteq X$ to be open if and only if its preimage

$$\pi^{-1}(U) = \{y \in Y : [y] \in U\}$$

is open in Y.

We now make a few further definitions that readers are invited to compare against the definitions they are familiar with from their experience with metric spaces.

Definition 0.36 (Closed subsets)**.** If X is a topological space, a subset F is *closed* if its complement U is *open*.

Definition 0.37 (Continuous functions)**.** If X and Y are topological spaces, a function $f : X \to Y$ is *continuous* if $f^{-1}(U)$ is an open subset of X whenever U is an open subset of Y.

Example 0.38. If Y is a topological space, $\sim$ an equivalence relation on Y, and $X = Y/\sim$ the quotient space as in example 0.35 above, then the map $\pi : Y \to X$ which carries a point $y \in Y$ to its equivalence class is continuous. This follows immediately from the definition of the quotient topology.

Definition 0.39 (Homeomorphisms)**.** Suppose X and Y are topological spaces and $f : X \to Y$ is a function. Then f is a *homeomorphism* if it is bijective, continuous, and $f^{-1} : Y \to X$ is continuous. More generally, f is a *homeomorphism onto its image* if is injective, continuous, and $f^{-1} : f(X) \to X$ is also continuous.

Definition 0.40 (Subspace)**.** If X is a topological space and S is a subset, then the *subspace topology* on S is

$$\tau_S := \{S \cap U : U \in \tau_X\}.$$

Further, S equipped with the subspace topology is called a *subspace* of X.

Exercise 0.41. If X is a topological space and S is a subspace, prove that the inclusion map $i : S \to X$ is continuous.

0.D.2. Hausdorff spaces

Definition 0.42. A topological space X is *Hausdorff* if, for every pair of distinct points $x, y \in X$, there exist disjoint open subsets U and V containing x and y, respectively.

Example 0.43. If X is a metric space, the corresponding topological space as in example 0.34 is automatically Hausdorff.

Exercise 0.44. Quotient spaces need not be Hausdorff. For example, let $Y = \mathbb{R} \times \{\pm 1\}$ inside $\mathbb{R}^2$, and define an equivalence relation $\sim$ by declaring that $(a, 1) \sim (a, -1)$ for all $a \neq 0$. Show that $X = Y/\sim$ is not Hausdorff.

0.D.3. Compact and σ-compact spaces

Definition 0.45 (Open covers)**.** An *open cover* of a topological space X is a collection $\mathcal{U}$ of open subsets of X which cover X, in the sense that

$$\bigcup_{U \in \mathcal{U}} U = X.$$

A *subcover* of $\mathcal{U}$ is a subset $\mathcal{U}' \subseteq \mathcal{U}$ which is itself an open cover.

Definition 0.46 (Compact)**.** A topological space X is *compact* if every open subcover has a finite subcover.

The following may be notions that you have not seen before in the context of metric spaces.

Definition 0.47. A topological space X is *σ-compact* if it is a countable union of compact subspaces.

Example 0.48. $\mathbb{R}^n$ is σ-compact, since

$$\mathbb{R}^n = \bigcup_{k=1}^{\infty} [-k, k]^n$$

and each hypercube $[-k, k]^n$ is compact.

1. Single variable derivatives

The goal in this chapter is to formalize derivatives of functions whose input is a single real number and whose output is also a single real number.

1.A. Two definitions of the derivative

1.A.1. Slope of the tangent line

Let S be a subset of $\mathbb{R}$ and $a \in S$ an interior point. We would like to define the tangent line to the graph of a function $f : S \to \mathbb{R}$ at a. We know intuitively what "tangent line" should mean: a line that "just barely touches" the graph of f at the point $(a, f(a))$. But it's not immediately clear how to formalize this intuition.

One approach to formalization is to notice that our intuitive idea of "tangent line" can be approximated by secant lines, and secant lines can be formalized without ambiguity. More precisely, for small values of h, the point $a + h$ is close to a and still in S since a is an interior point of S. The slope of the secant line passing through the two points $(a, f(a))$ and $(a + h, f(a + h))$ is computed by the quantity

$$\frac{f(a+h) - f(a)}{h}.$$

This quantity is often called a *difference quotient*. See figure 1.1 for a geometric explanation of this difference quotient. By letting $h \to 0$, we get closer and closer to what we intuitively understand to be the slope of the tangent line.

Definition 1.2. A function $f : S \to \mathbb{R}$ is *differentiable* at an interior point $a \in S$ if

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists. If this limit does exist, its value is called the *derivative of* f *at* a, and it is denoted either $f'(a)$ or

$$\left.\frac{df}{dx}\right|_{x=a}.$$

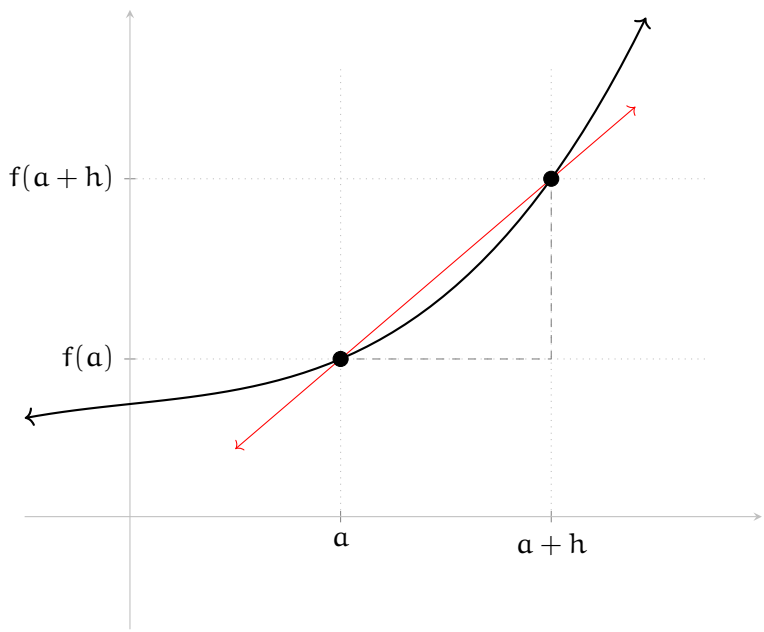

Figure 1.1.: The graph of a function f is depicted in black. The secant line passing through the two points $(a, f(a))$ and $(a+h, f(a+h))$ is depicted in red. The "run" between these points, is h. The "rise" between these points is $f(a+h) - f(a)$. Thus the slope of the secant line, computed as "rise over run," is precisely the difference quotient $(f(a+h) - f(a))/h$.

Exercise 1.3. Use the definition of the derivative to determine whether or not each of the following functions f is differentiable at the given point a. If it is differentiable, calculate $f'(a)$. Check your answers by making sure they agree with geometric intuition and/or rules you learned during your introductory calculus course.

(a) $f(x) = |x|$ at $a = 0$.

(b) $f(x) = x$ at $a = 0$.

(c) $f(x) = x^2$ at $a = 0$.

(d) $f(x) = x^2$ at $a = 2$.

(e) $f(x) = \begin{cases} 1 & x \geqslant 0 \\ 0 & x < 0 \end{cases}$ at $a = 0$.

Exercise 1.4 (Power rule for positive integer exponents)**.** Prove that, if $f(x) = x^n$ for a positive integer n, then $f'(a) = na^{n-1}$. *Hint.* Recall the binomial theorem, which says that

$$(a+h)^n = \sum_{k=0}^{n} \binom{n}{k} a^k h^{n-k}.$$

Exercise 1.5 (Differentiability implies continuity)**.** Prove that if $f : U \to \mathbb{R}$ is differentiable at $a \in U$, then f is also continuous at a.

1.A.2. Best linear approximation

Before proceeding further, we reinterpret the definition of the derivative in a way that will be better suited to generalization to the multivariable setting. Suppose a function $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$. The tangent line to the graph at the point $(a, f(a))$ is the graph of the function

$$t(x) = f'(a)(x - a) + f(a).$$

Since this function t is "close" to f near a, we should have that

$$f(a+h) - f(a) \approx t(a+h) - t(a) = f'(a)h$$

for small values of h. See figure 1.6.

Notice that the function $h \mapsto f'(a)h$ is a linear map $\mathbb{R} \to \mathbb{R}$ in the sense of linear algebra. One way to interpret the following result is that the function $h \mapsto f'(a)h$ is the *only* linear map that is a good approximation to $h \mapsto f(a+h) - f(a)$. The precise statement of the result uses little-oh notation; if you have not seen this notation before, see section 0.A.

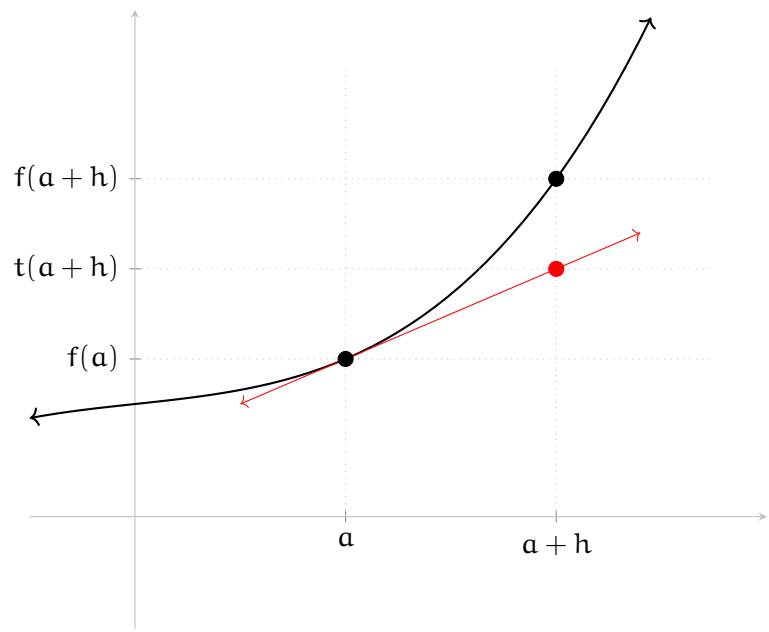

Figure 1.6.: The graph of a function f is depicted in black, and its tangent line t at a is depicted in red. Then the vertical distance $f(a+h) - f(a)$ is approximated by the vertical distance $t(a+h) - t(a)$. Since the tangent line has slope $f'(a)$, this "rise" can be computed as $f'(a)$ times the "run" (in other words, $f'(a)h$). As h gets smaller, this approximation gets better.

Proposition 1.7. *A function* $f : S \to \mathbb{R}$ *is differentiable at an interior point* $a \in S$ *if and only if there exists a linear map* $\ell : \mathbb{R} \to \mathbb{R}$ *such that*

$$|f(a+h) - f(a) - \ell(h)| = o(|h|) \text{ as } h \to 0.$$

Moreover, if there exists such a linear map ℓ*, then* ℓ *is uniquely determined: it must be given by*

$$\ell(h) = f'(a)h$$

for all $h \in \mathbb{R}$*.*

Proof. Let ℓ be a linear map $\ell(h) = ch$, where c is some scalar. Observe that

$$\begin{aligned}\frac{|f(a+h) - f(a) - \ell(h)|}{|h|} &= \left|\frac{f(a+h) - f(a) - ch}{h}\right| \\ &= \left|\frac{f(a+h) - f(a)}{h} - c\right|.\end{aligned}$$

This leads to the following sequence of "if and only if" statements.

$$\begin{aligned}|f(a+h) - f(a) - \ell(h)| = o(|h|) &\iff \lim_{h\to 0} \frac{|f(a+h) - f(a) - \ell(h)|}{|h|} = 0 \\ &\iff \lim_{h\to 0} \left|\frac{f(a+h) - f(a)}{h} - c\right| = 0 \\ &\iff \lim_{h\to 0} \left(\frac{f(a+h) - f(a)}{h} - c\right) = 0 \\ &\iff \lim_{h\to 0} \frac{f(a+h) - f(a)}{h} = c\end{aligned}$$

The statement of the proposition follows. □

Definition 1.8. If $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in U$, we define $df_a : \mathbb{R} \to \mathbb{R}$ to be the linear map $df_a(h) = f'(a)h$.

In other words, df_a is the only good linear approximation to the function

$$h \mapsto f(a+h) - f(a).$$

Being the *only good* linear approximation, it is also the *best* linear approximation, whence the name of this section.

1.B. Rules of differentiation

The goal in this section is to prove the rules for differentiation that one normally encounters in an introductory calculus course.

1.B.1. Sum and scalar multiples rule

The following can be proved using either the definition of the derivative 1.2 or the equivalent "best linear approximation" characterization of proposition 1.7. My suggestion is that you try both methods for these exercises. When you use proposition 1.7, you may find exercise 0.8 useful.

Exercise 1.9 (Constant functions rule)**.** Prove that, if $f : S \to \mathbb{R}$ is constant (ie, there exists a real number c such that $f(x) = c$ for all $x \in S$), then $f'(a) = 0$ for all interior points $a \in S$.

Exercise 1.10 (Sum rule)**.** Prove that, if $f, g : S \to \mathbb{R}$ are both differentiable at an interior point $a \in U$, then $f + g$ is also differentiable at a and

$$(f + g)'(a) = f'(a) + g'(a).$$

Exercise 1.11 (Scalar multiples rule)**.** Prove that, if c is a constant and $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$, then cf is also differentiable at a and

$$(cf)'(a) = cf'(a).$$

1.B.2. Product and quotient rules

Proposition 1.12 (Product rule)**.** *Suppose* $f, g : U \to \mathbb{R}$ *are both differentiable at* $a \in U$*. Then* fg *is also differentiable at* a *and*

$$(fg)'(a) = f'(a)g(a) + f(a)g'(a).$$

Proof. The proof is a clever algebraic manipulation of difference quotients. The key trick is introducing a "cross term" of the form $f(a)g(a + h)$, and then cancelling it out by also introducing its negative. More precisely, for any sufficiently small real number h, we have

$$\begin{aligned}
\frac{(fg)(a+h) - (fg)(a)}{h} &= \frac{f(a+h)g(a+h) - f(a)g(a)}{h} \\
&= \frac{f(a+h)g(a+h) - f(a)g(a+h) + f(a)g(a+h) - f(a)g(a)}{h} \\
&= \frac{f(a+h)g(a+h) - f(a)g(a+h)}{h} + \frac{f(a)g(a+h) - f(a)g(a)}{h} \\
&= \frac{f(a+h) - f(a)}{h} \cdot g(a+h) + f(a) \cdot \frac{g(a+h) - g(a)}{h}.
\end{aligned}$$

Taking the limit as $h \to 0$ yields the result. $\square$

Exercise 1.13. Check your understanding of the above proof of the product rule.

(a) Could you have instead introduced a cross term of the form $f(a+h)g(a)$? If so, rewrite the proof using this cross term. If not, explain why not.

(b) At what point is exercise 1.5 tacitly used in the above proof?

Exercise 1.14. Reprove the scalar multiples rule (exercise 1.11) using the product rule.

Exercise 1.15 (Quotient rule)**.** Suppose $f, g : S \to \mathbb{R}$ are both differentiable at an interior point $a \in U$, and that $g(a) \neq 0$. Prove that f/g is also differentiable at a and

$$(f/g)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{g(a)^2}.$$

Hint. Observe that

$$\frac{(f/g)(a+h) - (f/g)(a)}{h} = \frac{\frac{f(a+h)}{g(a+h)} - \frac{f(a)}{g(a)}}{h}.$$

Multiply both the numerator and denominator by $g(a+h)g(a)$ to clear the denominators inside the numerator, and then introduce the cross term $f(a)g(a)$. Be sure to notice at what point you use exercise 1.5.

Exercise 1.16 (Power rule for integer exponents)**.** Extend the result of exercise 1.4 to arbitrary integer exponents. *Hint.* Use the quotient rule for negative integers. The exponent zero is a special case; deal with it separately.

1.B.3. Chain rule

Proposition 1.17 (Chain rule)**.** *Suppose that S and T are both subsets of $\mathbb{R}$, that $f : S \to T$ is differentiable at an interior point a, that $f(a)$ is an interior point of T, and that $g : T \to \mathbb{R}$ is differentiable at $f(a)$. Then the composite $g \circ f : S \to \mathbb{R}$ is also differentiable at a, and*

$$(g \circ f)'(a) = g'(f(a))f'(a).$$

Let us discuss two different (but similar) proofs of this important result.

1.B.3.A. First proof

The first proof uses the definition of the derivative 1.2, but there is a subtle technicality involved; so, let us discuss it informally first before diving into the formal proof.

The idea involved is again a clever rewriting of difference quotients. Again, we introduce a cross term, which in this is $f(a+h) - f(a)$. But this time we introduce this cross term "multiplicatively" rather than "additively." More precsiely, we have

$$\begin{aligned}\frac{(g \circ f)(a+h) - (g \circ f)(a)}{h} &= \frac{g(f(a+h)) - g(f(a))}{h} \\ &= \frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} \cdot \frac{f(a+h) - f(a)}{h}.\end{aligned}$$

The difference quotient on the right is precisely $f'(a)$. Thus, it would be sufficient to prove

$$\lim_{h \to 0} \frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} = g'(f(a)). \tag{1.18}$$

This formula looks an awful lot like the difference quotient that is used to define the derivative of g at $f(a)$. In fact, since f is continuous at a by exercise 1.5, we have $f(a+h) \approx f(a) + h$ for small values of h, which means that

$$\frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} \approx \frac{g(f(a)+h) - g(f(a))}{f(a)+h-f(a)} = \frac{(g \circ f)(a+h) - (g \circ f)(a)}{h}$$

for small values of h. Taking the limit as $h \to 0$ on the right-hand side yields precisely $g'(f(a))$, so equation (1.18) seems like a reasonable conjecture.

The problem is that equation (1.18) isn't true. Issues arise when $f(a+h) = f(a)$ for arbitrarily small values of h, which leads to an infinite sequence of divisions by 0.

Example 1.19. Let

$$f(x) = \begin{cases} x^2 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

and let $g(x) = x^2$. Observe that $g'(f(0)) = 0$. You will show later (in exercise 1.30) that f is in fact differentiable at 0, so these functions do satisfy the hypothesis of the chain rule. However, the limit as $h \to 0$ of

$$\frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} = \frac{(h^2 \sin(1/h))^2}{h^2 \sin(1/h)}$$

does not exist. For example, the sequence $h_k = 1/k\pi$ converges to 0 as $k \to \infty$, but the expression above is undefined along this sequence.

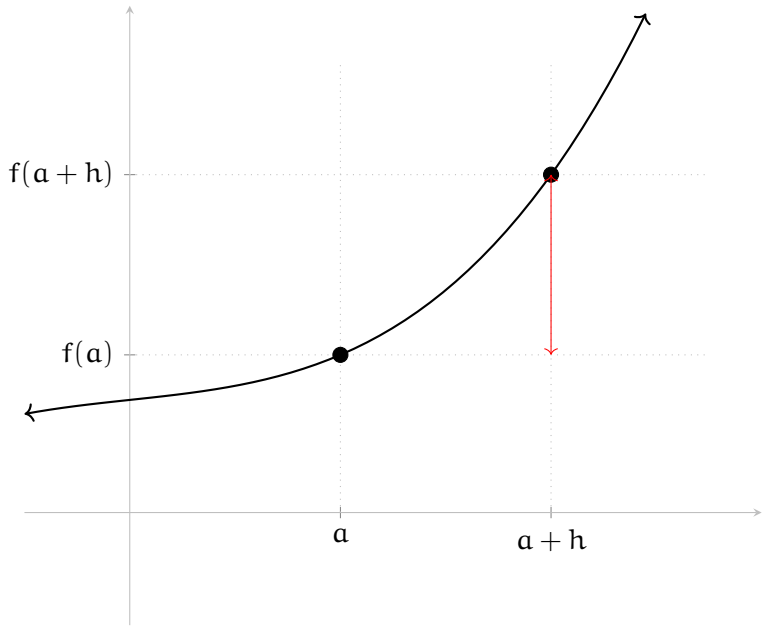

Figure 1.21.: The graph of f is depicted in black. The value of the function t on a small nonzero input value h is the vertical distance depicted in red.

We will get around the kind of issue that is described by the above example by carefully studying the expression

$$\frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)}. \tag{1.20}$$

Observe that, when $f(a+h) \neq f(a)$, then this expression is precisely the difference quotient that calculates the slope of the secant line passing through $(f(a), g(f(a))$ and $(f(a+h), g(f(a+h))$. When $f(a+h) = f(a)$, this secant line doesn't make sense; but what does make sense is the tangent line! So we'll remove the undefinedness of expression (1.20) by extending it so that it takes the value $g'(f(a))$ whenever $f(a+h) = f(a)$.

This leads us to our first proof of the chain rule.

First proof of the chain rule. Consider the function ϵ defined for small values of h by $\epsilon(h) = f(a+h) - f(a)$. See figure 1.21. You will verify in exercise 2.11 below that ϵ is continuous at 0.

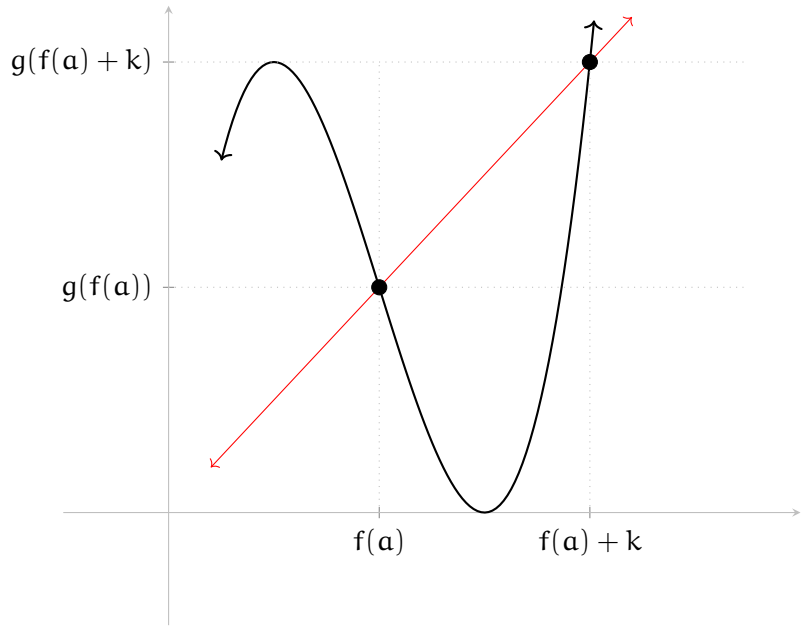

Figure 1.22.: The graph of g is depicted in black. The output of the function σ on a small nonzero input value k is the slope of the red secant line passing through the two points $(f(a), g(f(a))$ and $(f(a)+k, g(f(a)+k))$.

Next, define a function σ by the following.

$$\sigma(k) = \begin{cases} \dfrac{g(f(a)+k) - g(f(a))}{k} & \text{if } k \neq 0 \\ g'(f(a)) & \text{if } k = 0 \end{cases}$$

In other words, σ is the function that, on some small nonzero input k, outputs the slope of the secant line connecting the two points on the graph of g where $x = f(a)$ and $x = f(a)+k$. See figure 1.22. It follows from the definition of the derivative of g at $f(a)$ that the function σ is continuous at 0.

Observe that

$$\frac{(g \circ f)(a+h) - (g \circ f)(a)}{h} = \sigma(\epsilon(h)) \cdot \frac{f(a+h) - f(a)}{h} \tag{1.23}$$

for all small values of h, as you will verify in exercise 2.11 below. Thus

$$\begin{aligned}\lim_{h\to 0}\frac{(g\circ f)(a+h)-(g\circ f)(a)}{h} &= \lim_{h\to 0}\sigma(\epsilon(h))\cdot\frac{f(a+h)-f(a)}{h}\\ &\overset{(*)}{=} \sigma(\epsilon(0))f'(a)\\ &= g'(f(a))f'(a)\end{aligned}$$

where we have used the continuity of ϵ and σ and the definition of the derivative of f at a for the step labeled $(*)$ above. This proves the chain rule. □

Exercise 1.24. Verify the missing details in the above proof. Specifically, check the following.

(a) Prove that ϵ is continuous at 0.

(b) Prove equation (1.23). *Hint.* As suggested by our analysis before the proof of the chain rule, split up your proof of equation (1.23) into two cases: when $f(a+h)\neq f(a)$, and when $f(a+h)=f(a)$.

1.B.3.B. Second proof

Our second proof of the chain rule will use the "best linear approximation" interpretation of derivatives provided by proposition 1.7.

Second proof of the chain rule. By proposition 1.7, it is sufficient to prove that

$$g(f(a+h))-g(f(a))-dg_{f(a)}(df_a(h))=o(|h|) \text{ as } h\to 0.$$

Define the following "error in approximation" functions.

$$\begin{aligned}r(h) &= f(a+h)-f(a)-df_a(h)\\ s(k) &= g(f(a)+k)-g(f(a))-dg_{f(a)}(k)\end{aligned}$$

See figure 1.25.

Since

$$f(a+h)=f(a)+df_a(h)+r(h)$$

by definition of r, we have

$$\begin{aligned}g(f(a+h)) &= g(f(a)+df_a(h)+r(h))\\ &= g(f(a))+dg_{f(a)}(df_a(h)+r(h))+s(df_a(h)+r(h))\\ &= g(f(a))+dg_{f(a)}(df_a(h))+dg_{f(a)}(r(h))+s(df_a(h)+r(h))\end{aligned}$$

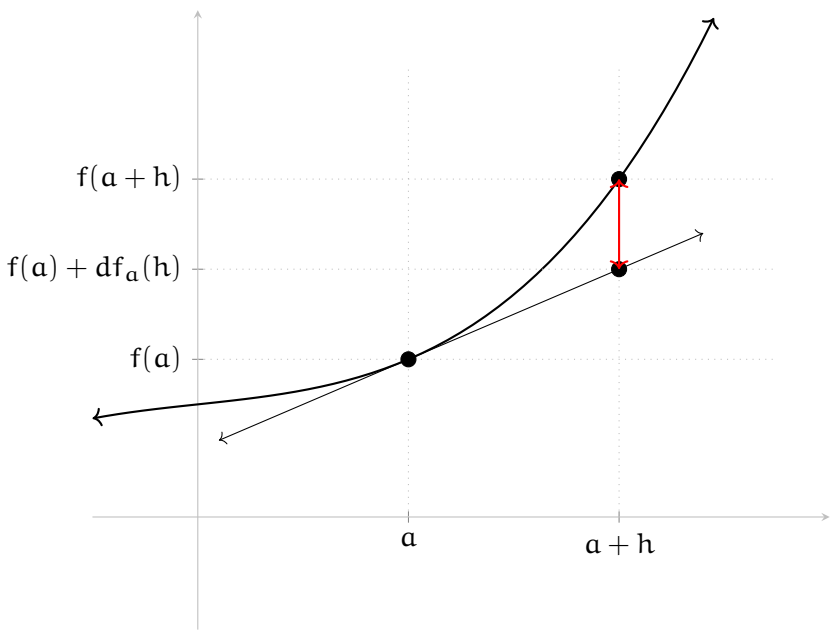

Figure 1.25.: The thick black curve is the graph of a function f and the thinner black line is its tangent line $f(a) + df_a$ are depicted in black. On a small input h, the "error in approximation" function r outputs the vertical distance indicated in red.

where we use the definition of s on input $k = df_a(h) + r(h)$ for the second step, and the fact that $dg_{f(a)}$ is linear for the third. Thus

$$g(f(a+h)) - g(f(a)) - dg_{f(a)}(df_a(h)) = dg_{f(a)}(r(h)) + s(df_a(h) + r(h)).$$

We therefore want to prove that the right-hand side is $o(|h|)$.

Since

$$dg_{f(a)}(r(h)) = g'(f(a)) \cdot r(h)$$

is just a scalar multiple of r, and $r = o(|h|)$, we see that $dg_{f(a)}(r) = o(|h|)$ also. So it is sufficient to prove that

$$s(df_a(h) + r(h)) = o(|h|).$$

We've now hit the technical part, so let's hit pause.

To see why this is technical, and for essentially the same reasons as the previous proof, notice that we're trying to prove that

$$\lim_{h \to 0} \frac{|s(df_a(h) + r(h)|)}{|h|} = 0.$$

To prove this, it's tempting to introduce a multiplicative cross-term as follows.

$$\frac{|s(df_a(h) + r(h)|)}{|h|} = \frac{|s(df_a(h) + r(h)|)}{|df_a(h) + r(h)|} \cdot \frac{|df_a(h) + r(h)|}{|h|}$$

But notice that $df_a(h) + r(h) = f(a+h) - f(a)$ by definition of r, so this cross term is problematic for the same reasons as before! If $f(a+h) = f(a)$ for h arbitrary close to 0, we have an infinite sequence of divisions by zeroes that causes problems.

We'll deal with this problem in essentially the same way as before: namely, we extend the definition of

$$\frac{|s(df_a(h) + r(h)|)}{|df_a(h) + r(h)|}$$

so that it is also defined when $f(a+h) = f(a)$.

Second proof of the chain rule, continued. Define $\eta(k) = s(k)/k$. Observe that $\eta(0) = 0$. Moreover, since $s(k) = o(|k|)$ by proposition 1.7, we see that η is continuous at 0. Then

$$\begin{aligned}\frac{|s(df_a(h) + r(h))|}{|h|} &= \frac{|\eta(df_a(h) + r(h))| \cdot |df_a(h) + r(h)|}{|h|} \\ &\leqslant |\eta(df_a(h) + r(h))| \cdot \left(|f'(a)| + \frac{|r(h)|}{|h|}\right).\end{aligned}$$

Observe that $df_a + r$ is continuous at 0 by exercise 1.26 below; moreover, we have just seen that η is continuous at $(df_a + r)(0) = 0$. Thus the composite $\eta \circ (df_a + r)$ is also continuous at 0. Thus

$$\lim_{h \to 0} |\eta(df_a(h) + r(h))| \cdot \left(|f'(a)| + \frac{|r(h)|}{|h|} \right) = 0,$$

where we have used the fact that $|r(h)| = o(|h|)$ to see that the parenthetical expression has a finite limit (namely, $|f'(a)|$). So, by the squeeze theorem, we can therefore conclude that $|s(df_a(h) + r(h))| = o(|h|)$. □

Exercise 1.26. Check that the function r is continuous at 0.

Notice that in the first proof, the trick to overcoming the technicality was defining the function σ at $k = 0$. In the second proof, the trick to overcoming the same technicality was defining η at $k = 0$. These two tricks are really the same trick, as shown by the following.

Exercise 1.27. Prove that the function η from the second proof and the function σ from the first proof are related by the equation

$$\sigma(k) = g'(f(a)) + \eta(k).$$

1.B.3.C. Applications of the chain rule

Exercise 1.28 (Derivatives of inverses)**.** Suppose S and T are both subsets of $\mathbb{R}$ and $f : S \to T$ is bijective, with inverse function $f^{-1} : T \to S$. Suppose further that f is differentiable at an interior point $a \in S$, that $f'(a) \neq 0$, and that $b = f(a)$ is an interior point of T. Prove that f^{-1} is differentiable at b and

$$(f^{-1})'(b) = \frac{1}{f'(a)} = \frac{1}{f'(f^{-1}(b))}.$$

Exercise 1.29 (Power rule for fractional exponents)**.** Extend the result of exercise 1.16 to arbitrary rational exponents (ie, to exponents that can be written in the form m/n where m and n are both integers).

1.B.4. Some rules without proofs

Here are some other rules you probably remember from calculus. We won't prove them here, but you should feel free to use them.

(1) If $f(x) = e^x$, then $f'(x) = e^x$.

(2) If $f(x) = \ln(x)$, then $f'(x) = 1/x$.

(3) If $f(x) = \sin(x)$, then $f'(x) = \cos(x)$.

(4) If $f(x) = \cos(x)$, then $f'(x) = -\sin(x)$.

(5) If $f(x) = x^r$ for some *real* number r, then $f'(x) = rx^{r-1}$.

Note that by doing exercise 1.29, you have already proved the most common cases of the power rule (5).

The proofs of these facts are omitted not because the proofs are particularly difficult, but because we would have to go on a long detour to even define the functions involved sufficiently rigorously to prove anything about them. That would distract us from our object of meditation: the derivative.

Exercise 1.30. (a) Prove that the function

$$f(x) = \begin{cases} x\sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is continuous at 0 but is not differentiable at 0.

(b) Prove that the function

$$f(x) = \begin{cases} x^2\sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is differentiable at 0 but the derivative f' is not continuous at 0.

1.C. Derivative at local extrema

We first recall the following definition.

Definition 1.31. If X is a metric space and $f : X \to \mathbb{R}$ is a function, a point $a \in X$ is said to be a *local maximum* of f if there exists a neighborhood U of a in X such that $f(x) \leqslant f(a)$ for all $x \in U$. We define *local minimum* by reversing the inequality. A *local extremum* is a point that is either a local maximum or a local minimum.

The following is a key property of derivatives.

Exercise 1.32 (Fermat's interior extremum theorem)**.** Suppose an interior point $a \in S$ is a local extremum of a function $f : S \to \mathbb{R}$. If f is differentiable at a, prove that $f'(a) = 0$. *Hint*. Consider the following function (cf. the first proof of the chain rule in section 1.B.3.A).

$$\sigma(h) = \begin{cases} \dfrac{f(a+h) - f(a)}{h} & \text{if } h \neq 0 \\ f'(a) & \text{if } h = 0 \end{cases}$$

Since f is differentiable at $x = a$, σ is continuous at $h = 0$. If a is a local minimum, what can you say about the values of σ for nonzero values of h? What about if a is a local maximum?

1.D. Differentiable functions

Definition 1.33. If U is an open subset of $\mathbb{R}$, a function $f : U \to \mathbb{R}$ is *differentiable* if f is differentiable at every $a \in U$. The function $U \to \mathbb{R}$ defined by $a \mapsto f'(a)$ is called the *derivative* of f, and is denoted by f'.

1.D.1. Mean value theorem

The mean value theorem 1.35 is an important foundational result; it allows us to prove many results about derivatives which we are familiar with from calculus. Rolle's theorem 1.34 is a special case of the mean value theorem, but the general case can be derived from this special case.

Exercise 1.34 (Rolle's theorem)**.** Suppose $a < b$ and let $f : [a, b] \to \mathbb{R}$ be a continuous function that is differentiable on (a, b). If $f(a) = f(b) = 0$, prove that there exists $c \in (a, b)$ such that $f'(c) = 0$. *Hint*. Any function must fall into at least one of the following three cases: (i) f is constantly zero, (ii) f takes on positive values somewhere on $[a, b]$, and (iii) f takes on negative values somewhere on $[a, b]$. Deal with the first case using exercise 1.9, and the latter two cases using the extreme value theorem[1] together with exercise 1.32.

Exercise 1.35 (Mean value theorem)**.** Suppose $a < b$ and let $f : [a, b] \to \mathbb{R}$ be a continuous function that is differentiable on (a, b). Prove that there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Hint. Let ℓ be the secant line passing through $(a, f(a))$ and $(b, f(b))$. Then consider the function $g = f - \ell$. See figure 1.36. Then $g(a) = g(b) = 0$, so we can apply Rolle's theorem

[1] The *extreme value theorem* says that if K is a compact metric space and $f : K \to \mathbb{R}$ is continuous, then f achieves its global maximum and its global minimum somewhere on K.

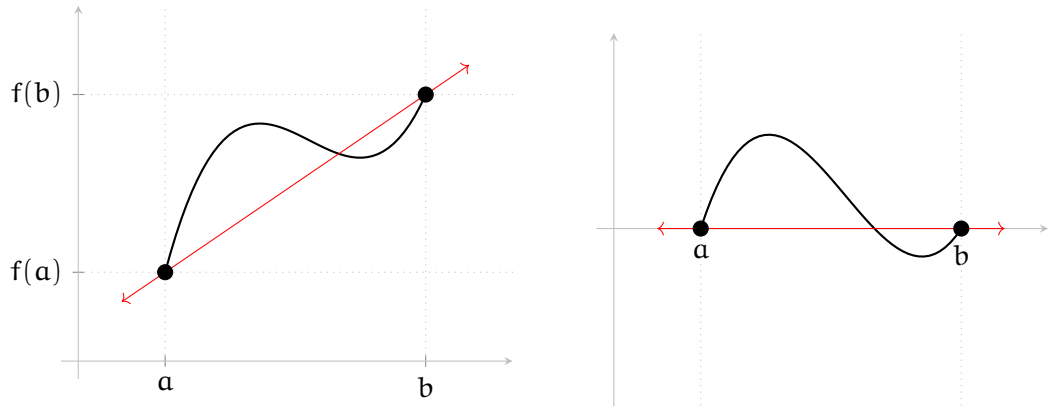

Figure 1.36.: On the left is depicted the graph of a function f and the secant line ℓ. Subtracting ℓ from the picture on the left yields the picture on the right.

1.34 to g.

1.D.2. Consequences of the mean value theorem

Here is a standard property of derivatives from calculus; the proof follows quickly from the mean value theorem.

Lemma 1.37. *Suppose I is an open interval and that $f : I \to \mathbb{R}$ is differentiable. If $f' = 0$, then f is constant.*

Proof. We need to show that, for any distinct pair of points $a, b \in I$, we have $f(a) = f(b)$. We can assume without loss of generality that $a < b$. Since f is differentiable on I and $[a, b] \subseteq I$, we know in particular that f is continuous on $[a, b]$ and differentiable on (a, b), so the mean value theorem applies: there exists a point $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$

But $f'(c) = 0$ by assumption, so $f(a) = f(b)$. □

Next, we recall the following definition.

Definition 1.38. Suppose $f : S \to \mathbb{R}$ is a function and $x, y \in S$.

- f is *increasing* if $x < y$ implies $f(x) \leqslant f(y)$.

- f is *decreasing* if $x < y$ implies $f(x) \geqslant f(y)$.
- f is *strictly increasing* if $x < y$ implies $f(x) < f(y)$.
- f is *strictly decreasing* if $x < y$ implies $f(x) > f(y)$.

Further, f is *monotonic* if it is either increasing or decreasing, and *strictly monotonic* if it is either strictly increasing or strictly decreasing.

The following standard property of differentiable functions again follows quickly from the mean value theorem.

Exercise 1.39. Suppose I is an open interval and that $f : I \to \mathbb{R}$ is differentiable. Prove the following four statements.

(a) If $f' \geqslant 0$, then f is increasing.

(b) If $f' > 0$, then f is strictly increasing.

(c) If $f' \leqslant 0$, then f is decreasing.

(d) If $f' < 0$, then f is strictly decreasing.

Hint. For all four, imitate the proof of lemma 1.37.

1.D.3. Darboux's theorem $*$

Darboux's theorem is not commonly discussed in elementary calculus, nor is it strictly necessary for what follows. But it is an interesting fact about derivatives, and is sometimes useful to know. Before getting to the statement of the theorem, it is convenient to make the following definition.

Definition 1.40. Suppose I is an interval and $f : I \to \mathbb{R}$ is a function. Then f *has the intermediate value property* if, for every $a < b$ in I and every $y \in \mathbb{R}$ strictly between $f(a)$ and $f(b)$, there exists $x \in (a, b)$ such that $f(x) = y$.

The intermediate value theorem asserts that every continuous function has the intermediate value property. Simple examples of discontinuous functions don't have the intermediate value property (cf. exercise 1.41 below). But, it turns out that some discontinuous functions do have the intermediate value property (cf. exercise 1.42 below).

Exercise 1.41. Let I be an interval and $f : I \to \mathbb{R}$ a function.

(a) Recall that f has a *removable discontinuity* at a point $a \in I$ if

$$\lim_{x \to a} f(x)$$

exists, but the value of this limit does not equal $f(a)$. Show that, if f has a removable discontinuity, then f does *not* have the intermediate value property.

(b) Recall that f has a *jump discontinuity* at a point $a \in I$ if the two one-sided limits

$$\lim_{x \to a^-} f(x) \text{ and } \lim_{x \to a^+} f(x)$$

both exist, but their values are not equal to each other. Show that, if f has a jump discontinuity, then f does *not* have the intermediate value property.

Exercise 1.42. Show that the function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is discontinuous at $x = 0$ but has the intermediate value property. *Hint*. First show that for any non-degenerate interval I containing 0, we have $f(I) = [-1, 1]$. Then explain how this implies the intermediate value property.

We know that the derivative f' of a differentiable function f need not be continuous (cf. exercise 1.30). Darboux's theorem asserts that, even though f' might not be continuous, it must still have the intermediate value property! This means, for example, that derivatives cannot have removable or jump discontinuities (cf. exercise 1.41).

Theorem 1.43 (Darboux)**.** *Suppose* I *is an open interval and* $f : I \to \mathbb{R}$ *is differentiable. Then* f' *has the intermediate value property.*

Proof. Suppose $a < b$ are two elements of I and m is a real number strictly between $f'(a)$ and $f'(b)$. We want to show that there exists $c \in (a, b)$ such that $f'(c) = m$. This statement somewhat resembles the statement of the mean value theorem 1.35, and indeed, our proof will also somewhat resemble the proof we gave of the mean value theorem.

Consider the differentiable function $g : I \to \mathbb{R}$ defined by $g(x) = f(x) - mx$. See figure 1.44. Observe that $f'(c) = m$ if and only if $g'(c) = 0$. Thus we would like to show that there exists $c \in (a, b)$ such that $g'(c) = 0$.

There are two cases two consider: either (i) $f'(a) < m < f'(b)$ or (ii) $f'(a) > m > f'(b)$. Suppose we are in case (i). By the extreme value theorem, g attains its minimum on the

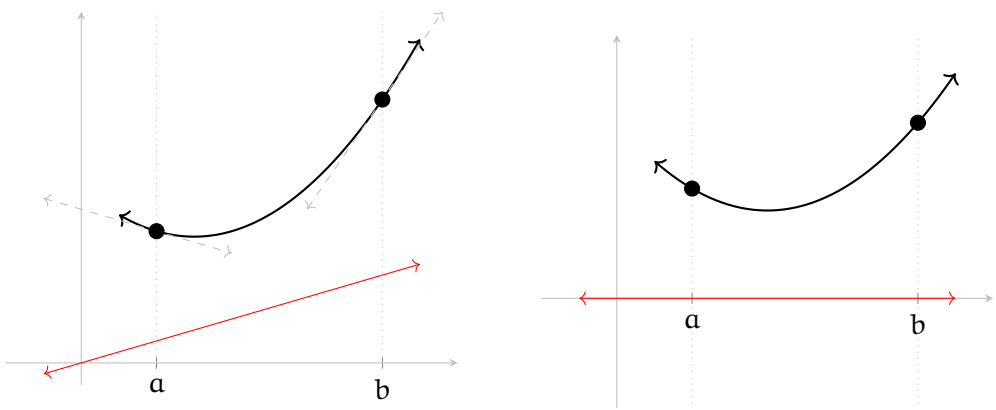

Figure 1.44.: On the left is depicted the graph of a function f, the two tangent lines at a and b, and the graph of the line $\ell(x) = mx$ for some m in between $f'(a)$ and $f'(b)$. Subtracting ℓ from this picture yields the picture on the right.

compact interval $[a, b]$ at some point $c \in [a, b]$. We claim that c must be an interior point of $[a, b]$, and then Fermat's interior extremum theorem 1.32 will imply that $g'(c) = 0$.

If $c = a$ were the minimum of g on $[a, b]$, then we would have

$$\frac{g(a+h) - g(a)}{h} > 0$$

for all sufficiently small positive values of h. This would mean that $g'(a) \geqslant 0$, but we know that $g'(a) = f'(a) - m < 0$, which is a contradiction. Similarly, if $c = b$ were the minimum of g on $[a, b]$, then we would have

$$\frac{g(b+h) - g(b)}{h} < 0$$

for all sufficiently small negative values of h. This would imply that $g'(b) \leqslant 0$, but we know that $g'(b) = f'(b) - m > 0$.

In case (ii), we instead let c be the *maximum* of g on $[a, b]$, and argue similarly to show that c must be an interior point. □

1.E. Levels of differentiability

1.E.1. C^n functions

Definition 1.45. Suppose U is an open subset of $\mathbb{R}$ and $f : U \to \mathbb{R}$ is differentiable. If the derivative $f' : U \to \mathbb{R}$ is also continuous, then f is said to be *continuously differentiable*.

Definition 1.46. Let U be an open subset of $\mathbb{R}$ and $f : U \to \mathbb{R}$ a function. We make the following string of definitions.

- f is C^0 if it is continuous.
- f is C^1 if it is continuously differentiable.
- f is C^2 if the derivative f' is C^1. In other words, this means that the derivative of the derivative $(f')'$, denoted f'' or $f^{(2)}$ and called the *second derivative*, exists and is continuous.
- In general, for any positive integer n, we say that f is C^n if f' exists and is C^{n-1}. This means that the n*th derivative* $f^{(n)} = (f^{(n-1)})'$ exists and is continuous.

Exercise 1.47. Prove that, for every non-negative integer n, the set of all C^n functions $U \to \mathbb{R}$ is a vector space.

Exercise 1.48 (General Leibniz Rule)**.** Suppose that $f, g : U \to \mathbb{R}$ are C^n functions. Prove that the product fg is also C^n, and that

$$(fg)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} f^{(k)} g^{(n-k)}.$$

Exercise 1.49. Show that the composite of two C^n functions is also C^n. *Hint*. Induct on n. The base case $n = 0$ is clear. For the inductive step, use the chain rule 1.17 and the first part of exercise 1.48.

Observe that we have an infinite chain of implications

$$\cdots \implies C^3 \implies C^2 \implies C^1 \implies C^0.$$

The following exercise shows that all of these implications are strict; in other words, for every n, there exists a C^n function which is not C^{n+1}.

Exercise 1.50. For every non-negative integer n, let

$$f_n(x) = x^n|x|.$$

Prove that f_n is C^n but not C^{n+1}. *Hint.* First prove that $f'_{n+1} = (n+2)f_n$.

1.E.2. Smooth functions

Definition 1.51. A function $f : U \to \mathbb{R}$ is said to be C^∞, or *infinitely differentiable*, or *smooth*, if it is in C^n for all n.

It's a little annoying that there are three different words that all mean the same thing, but that's just how it is; all three are commonly used in the mathematical literature, so it's best to get used to all of them. We'll usually use the word "smooth." It follows immediately from exercises 1.47 to 1.49 that sums, scalar multiples, products, and composites of smooth functions are also smooth.

An important principle is that we can tailor smooth functions to almost arbitrary specifications. Our first example of this principle is the following smooth bijection $(-1, 1) \to \mathbb{R}$ whose inverse is also smooth. A smooth bijection with a smooth inverse is also called a "diffeomorphism."

Example 1.52. Consider the function $f : (-1, 1) \to \mathbb{R}$ defined by

$$f(x) = \frac{x}{1 - x^2}.$$

See figure 1.53. Since this is a quotient of two smooth functions and the denominator is nonzero on $(-1, 1)$, this function is also smooth. We can calculate that

$$f'(x) = \frac{1 + x^2}{(1 - x^2)^2}$$

and we can see from this formula that the derivative is always strictly positive. Thus f is strictly increasing (cf. exercise 1.39); in partcular, it is injective. Moreover, we have

$$\lim_{x \to -1^+} f(x) = -\infty \quad \text{and} \quad \lim_{x \to 1^-} f(x) = +\infty$$

so the intermediate value theorem guarantees that f is bijective. Thus the inverse function $f^{-1} : \mathbb{R} \to (-1, 1)$ exists. Finally, since f' is always nonzero, it follows from lemma 1.54 that f^{-1} must be smooth. It is possible to verify that

$$f^{-1}(x) = \frac{-1 + \sqrt{4x^2 + 1}}{2x}$$

by check that composing this formula with f yields the identity, but, thanks to lemma 1.54, we don't actually need this formula for f^{-1} at all to know that f^{-1} is smooth.

Lemma 1.54. *Suppose U and V are open subsets of $\mathbb{R}$, that $f : U \to V$ is a smooth bijection, and that f' is always nonzero. Then f is a diffeomorphism.*

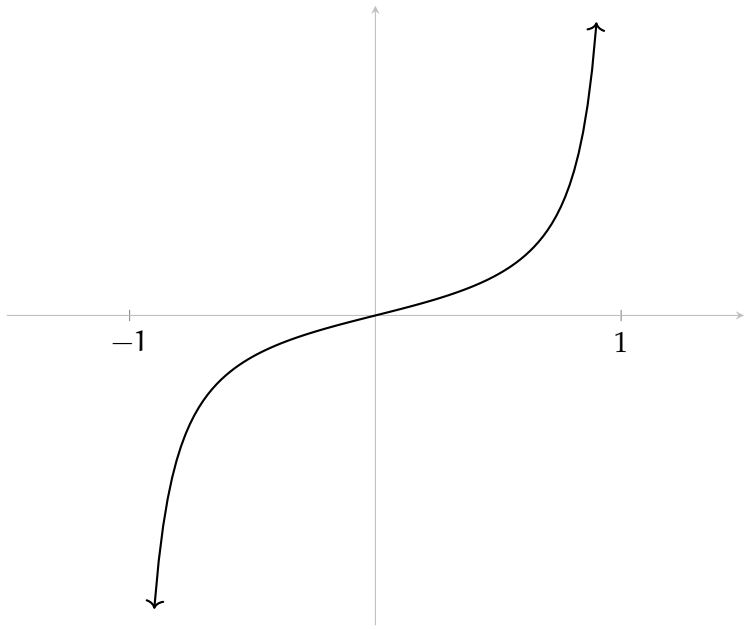

Figure 1.53.: The graph of the function f from example 1.52.

Proof. We want to show that $f^{-1} : V \to U$ is smooth. Since f' is always nonzero, we know from exercise 1.28 that f^{-1} is differentiable and that

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}.$$

In other words, $(f^{-1})'$ is the composite of three functions: f^{-1}, f', and inversion (ie, the function $x \mapsto 1/x$). Inversion is clearly smooth; since f is smooth, so is f'.

We now prove by induction that f^{-1} is C^n for all integers n. For the base case, we know from above that it is C^0 (in fact, we even know more: we know that f^{-1} is even differentiable, but all we need to know for the base case is that it is continuous). Suppose inductively that f^{-1} is C^n for some non-negative integer n. Then, as we saw above, $(f^{-1})'$ is the composite of f^{-1}, f', and inversion. The latter two are smooth, so in particular are C^n. By the inductive hypothesis, we know that f^{-1} is C^n. Thus $(f^{-1})'$, being the composite of three C^n functions, is also C^n by exercise 1.49. This means that f^{-1} is C^{n+1}, which completes the induction. □

Exercise 1.55. For any $a < b$, construct a diffeomorphism $f : (a, b) \to \mathbb{R}$.

Our next example of a slightly strange smooth function is the following one, which gets "infinitely flat" at 0 but is not constantly equal to 0.

Remark 1.56. You probably remember proving that

$$\lim_{u\to\infty} \frac{u^m}{e^u} = 0$$

for any non-negative integer m using L'Hôpital's rule in your calculus class. We haven't proved L'Hôpital's rule (too much of a tangent (pun intended)), but we'll use this fact anyway. In particular, note that by making the substitution $u = 1/x$ above, we find that

$$\lim_{x\to 0^+} \frac{e^{-1/x}}{x^m} = 0$$

for any non-negative integer m.

Exercise 1.57. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined as follows.

$$f(x) = \begin{cases} e^{-1/x} & \text{if } x > 0 \\ 0 & \text{if } x \leqslant 0 \end{cases}$$

See figure 1.58.

(a) Use remark 1.56 to show, using the definition of the derivative 1.2, that $f'(0) = 0$.

(b) Check that

$$f'(x) = \begin{cases} \dfrac{e^{-1/x}}{x^2} & \text{if } x > 0 \\ 0 & \text{if } x \leqslant 0. \end{cases}$$

You've already computed $f'(0)$. You can compute formulas for the derivative $f'(x)$ for $x > 0$ and for $x < 0$ using the usual rules for differentiation.

(c) Use remark 1.56 to show that $f''(0) = 0$, and then check that

$$f''(x) = \begin{cases} p_2(x) \cdot \dfrac{e^{-1/x}}{x^4} & \text{if } x > 0 \\ 0 & \text{if } x \leqslant 0 \end{cases}$$

where p_2 is the function $p_2(x) = 1 - 2x$.

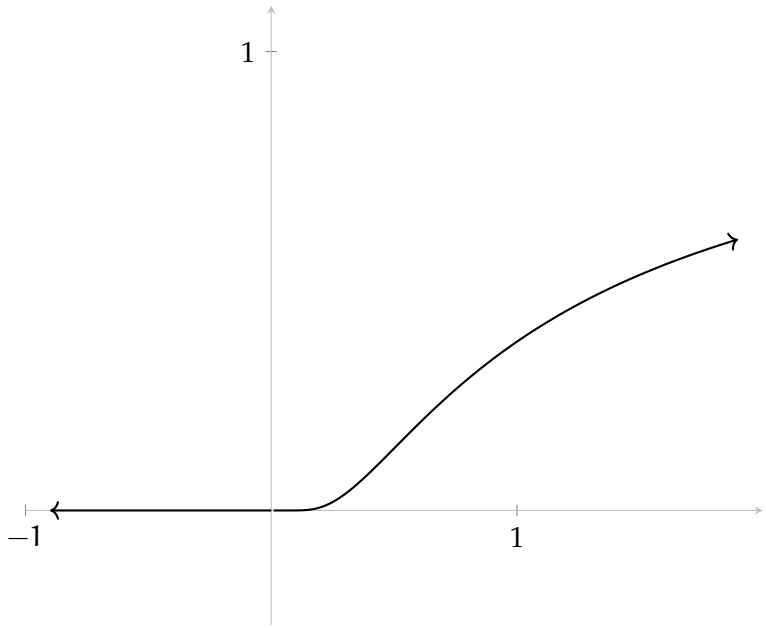

Figure 1.58.: The graph of the function f from exercise 1.57.

(d) Inductively, prove that

$$f^{(n)}(x) = \begin{cases} p_n(x) \cdot \dfrac{e^{-1/x}}{x^{2n}} & \text{if } x > 0 \\ 0 & \text{if } x \leqslant 0 \end{cases}$$

where p_n is a polynomial function of degree $n-1$ such that $p_n(0) = 1$.

Thus f is an example of a smooth function that is "infinitely flat" at $x = 0$, in the sense that $f^{(n)}(x) = 0$ for all n, but is not constantly equal to 0.

Next up, we have "bump functions." Recall that the "support" of a function $f : U \to \mathbb{R}$ is the closure of the set $\{x \in U : f(x) \neq 0\}$. A "bump function" is a smooth function whose support is compact.

Exercise 1.59. Consider the function $\psi : \mathbb{R} \to \mathbb{R}$ given by the following.

$$\psi(x) = \begin{cases} e^{-1/(1-x^2)} & \text{if } x \in (-1, 1) \\ 0 & \text{otherwise.} \end{cases}$$

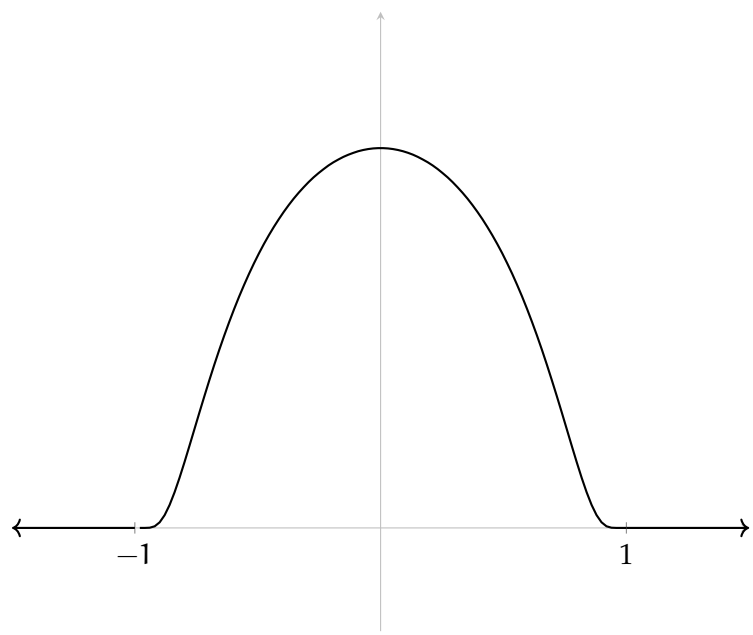

Figure 1.60.: The graph of the function ψ from exercise 1.59.

See figure 1.60 for its graph. Check that ψ is the composite of the "infinitely flat" function from exercise 1.57 and the polynomial $p(x) = 1 - x^2$. Conclude that ψ is a bump function whose support is $[-1, 1]$.

Exercise 1.61. For any $a < b$ in $\mathbb{R}$, construct a bump function whose support is $[a, b]$.

In the following two examples, we will have to invoke the fundamental theorem of calculus, even though we have not proved it (or even defined integrals, for that matter).

Example 1.62. Let ψ be the bump function from exercise 1.59, and then consider

$$\eta(x) = \int_{-1}^{x} \psi(t)\, dt.$$

Then

$$\eta'(x) = \frac{d}{dx} \int_{-1}^{x} \psi(t)\, dt = \psi(x)$$

by the fundamental theorem of calculus. Thus η is smooth, since $\eta' = \psi$ is smooth. Note moreover that η is increasing, since $\eta' = \psi \geqslant 0$. Finally, it is clear from the geometric interpretation of integrals as "area under the curve" that η is constantly 0 for all $x \leqslant -1$ and that it is constantly $\eta(1)$ for all $x \geqslant 1$.

Exercise 1.63. Let $I = [a, b]$ be a compact interval in $\mathbb{R}$ and let U be an open subset containing I. Prove that there exists a bump function $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = 1$ for all $x \in K$ and $f(x) = 0$ for all $x \notin U$. *Hint.* The characteristic function

$$\chi_I(x) = \begin{cases} 1 & \text{if } x \in I \\ 0 & \text{if } x \notin I \end{cases}$$

is close to having all of the properties we want, but it's not smooth (it's clearly not even continuous). Since U is open, there exist $a' < a$ and $b' > b$ such that $I := [a', b'] \subseteq U$. Modify χ_I on the intervals $[a', a]$ and $[b, b']$ to obtain the desired function. Use inspiration from the function η of example 1.62.

2. Multivariable derivatives

In this chapter, we will study derivatives of functions $S \to \mathbb{R}^n$ where $S \subseteq \mathbb{R}^m$. We dealt with the case $m = n = 1$ in the previous chapter, starting with a very geometric idea of tangent lines to graphs. For large values of m and n, graphs become hard to visualize and it is not so clear what "tangent" should mean. But there is at least one higher dimensional case when we can use some geometry to formalize what "tangent" should mean, which is the case $m = 2$ and $n = 1$. The graph of such a function is a surface in $\mathbb{R}^3$, and its "tangent" at a point should be a plane. We'll begin by analyzing a particular example of this situation to build up some geometric intuition, and then proceed with the abstract discussion of multivariable derivatives for general m and n.

2.A. An extended example

Let us begin with an example. Our analysis of this example will presage many of the concepts we will discuss later in the chapter. The example we'll consider is the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x_1, x_2) = x_1^2 + x_2^2.$$

2.A.1. Describing the graph of f

First off, let's try to get a solid understanding of the graph of f. The graph Γ is a subset of $\mathbb{R}^2 \times \mathbb{R} = \mathbb{R}^3$ defined by

$$\Gamma = \{(x_1, x_2, y) : y = f(x_1, x_2)\} = \{(x_1, x_2, y) : y = x_1^2 + x_2^2\}.$$

To visualize Γ, we will repeatedly slice Γ along various planes.

If we consider the slice obtained by setting $x_1 = 0$, we find a parabola $y = x_2^2$. Similarly, if we consider the slice obtained by setting $x_1 = 1$, we obtain the parabola $y = 1 + x_2^2$. Fixing $x_1 = 5$, we obtain the parabola $y = 25 + x^2$. In fact, we can see that no matter what value of x_1 we fix, the resulting slice is always a parabola, just translated up from the origin by differing amounts (that amount being x_1^2).

It's also worth considering the "level sets," which are the slices of Γ obtained by fixing various values of y. When $y = 0$, there is just a single point $x_1 = x_2 = 0$. Fixing $y = 1$, the slice is given by $1 = x_1^2 + x_2^2$, which is a circle of radius 1. Fixing $y = 17$, the slice is $17 = x_1^2 + x_2^2$, which is circle of radius $\sqrt{17}$. In fact, all of the level sets are circles.

Putting these slices together, we see that the graph Γ is a "big parabolic bowl."

2.A.2. Tangent plane at $a = (2, 1)$

Now that we understand what the graph of f looks like, let's try to figure out what the "tangent plane" T at a point $a = (2, 1)$ will look like. Of course, it's a plane inside $\mathbb{R}^3$ that passes through the point $(2, 1, f(2, 1)) = (2, 1, 5)$. To specify which plane it actually is, we start by describing some lines that lie on this plane.

Consider the $x_2 = 1$ slice. We know that the graph of Γ along this slice is the parabola $y = x_1^2 + 1$. So, slicing the tangent plane T along $x_2 = 1$ should yield the tangent line to this function of one variable, which we know how to compute from chapter 1. The slope of this tangent line is

$$\left.\frac{df(x_1, 1)}{dx_1}\right|_{x_1=2} = 4.$$

This quantity is called the *first partial derivative of* f, and will be denoted $\partial_1 f(a)$ later. Thus the tangent line is the line parametrized by

$$h_1 \mapsto (2, 1, 5) + h_1 \cdot (1, 0, 4).$$

Similarly, we can consider the $x_1 = 2$ slice. We know that the graph Γ along this slice is the parabola $y = 4 + x_2^2$.The slope of the corresponding tangent line is

$$\partial_2 f(a) = \left.\frac{df(2, x_2)}{dx_2}\right|_{x_2=1} = 2.$$

Thus this tangent line is the line parametrized by

$$h_2 \mapsto (2, 1, 5) + h_2 \cdot (0, 1, 2).$$

These two lines must uniquely determine the entire plane T. One way of describing T that immediately falls out of the parametrizations we've found of the tangent lines is the plane parametrized by

$$\begin{bmatrix} h_1 \\ h_2 \end{bmatrix} \mapsto \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix} + \begin{bmatrix} h_1 \\ h_2 \\ 4h_1 + 2h_2 \end{bmatrix}.$$

The important part of this expression is the bottom entry on the far right, the $4h_1 + 2h_2$. To see why, consider the function $\mathbb{R}^2 \to \mathbb{R}$ given by $(h_1, h_2) \mapsto 4h_1 + 2h_2$. This is what we will call the *(total) derivative* of f and denote by df_a. Notice that df_a is a linear map $\mathbb{R}^2 \to \mathbb{R}$. Moreover, its graph is a plane passing through the origin that is parallel to T. In other words, if we take the graph of df_a and translate it over to the point $(2, 1, 5)$, we obtain exactly the tangent plane T. Speaking more loosely, the derivative $df_a : \mathbb{R}^2 \to \mathbb{R}$ records all of the "slope-y information" about the tangent plane T.

Notice moreover that the standard matrix representation $[df_a]$ of df_a is

$$[df_a] = \begin{bmatrix} 4 & 2 \end{bmatrix} = \begin{bmatrix} \partial_1 f(a) & \partial_2 f(a) \end{bmatrix}. \tag{2.1}$$

2.A.3. df_a as an approximation

Consider the function $\epsilon : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$v \mapsto |f(a + v) - f(a) - df_a(v)|.$$

If $v = (h_1, h_2)$, we have

$$\epsilon(v) = \epsilon\left(\begin{bmatrix} h_1 \\ h_2 \end{bmatrix}\right) \mapsto |(2 + h_1)^2 + (1 + h_2)^2 - 5 - 4h_1 - 2h_2| = |h_1^2 + h_2^2| = \left|\begin{bmatrix} h_1 \\ h_2 \end{bmatrix}\right|^2 = |v|^2.$$

Notice that ϵ gets small rapidly as $v \to 0$. More precisely, notice that

$$\lim_{v \to 0} \frac{\epsilon(v)}{|v|} = \lim_{v \to 0} |v| = 0,$$

so $\epsilon = o(|v|)$ as $v \to 0$.

Since ϵ gets small rapidly, the function $v \mapsto f(a) + df_a(v)$ is a good approximation of the function $v \mapsto f(a + v)$ for small values of v. Said differently, letting $x = a + v$, the function $x \mapsto f(a) + df_a(x - a)$ is a good approximation of f near a.

2.A.4. Overview

In what follows, we will turn the above example on its head and *define* the derivative using approximations. In other words, the derivative df_a will be defined to be a linear function which yields a good approximation of f near a point a. Then we will prove in theorem 2.21 that the partial derivatives can be used to describe df_a, as we noted in equation (2.1) above. The reason for doing this is that there are some bizarre functions where partial derivatives make sense, but tangent planes do not (cf. exercise 2.19 below).

As you may be able to tell, we'll be using linear algebra heavily in this chapter. If you need a brief refresher, check out section 0.B. We'll also need some facts about the operator norm; you may not have seen this before, and can refer to section 0.C as needed.

2.B. Definition of the derivative

Throughout, S will denote a subset of $\mathbb{R}^m$.

Definition 2.2. A function $f : S \to \mathbb{R}^n$ is *differentiable at* an interior point $a \in S$ if there exists a linear map $\ell : \mathbb{R}^m \to \mathbb{R}^n$ such that

$$|f(a+v) - f(a) - \ell(v)| = o(|v|) \text{ as } v \to 0.$$

It turns out that there can exist at most one linear map that has the above property.

Lemma 2.3. *Suppose $f : S \to \mathbb{R}^m$ is differentiable at an interior point $a \in S$. If ℓ and ℓ' are both linear maps $\mathbb{R}^m \to \mathbb{R}^n$ such that*

$$|f(a+v) - f(a) - \ell(v)| = o(|v|) \textit{ and } |f(a+v) - f(a) - \ell'(v)| = o(|v|) \textit{ as } v \to 0,$$

then $\ell = \ell'$.

Proof. Let $\phi = \ell' - \ell$. Then ϕ is also a linear map $\mathbb{R}^m \to \mathbb{R}^n$. Moreover, observe that

$$\begin{aligned}
|\phi(v)| &= |\ell'(v) - \ell(v)| \\
&= |(f(a+v) - f(a) - \ell(v)) - (f(a+v) - f(a) - \ell'(v)| \\
&\leqslant |f(a+h) - f(a) - \ell(v)| + |f(a+v) - f(a) - \ell'(v)|.
\end{aligned}$$

Since both $|f(a+v) - f(a) - \ell(v)|$ and $|f(a+v) - f(a) - \ell'(v)|$ are $o(|v|)$, exercises 0.7 and 0.8 imply that $|\phi(v)| = o(|v|)$ also. Then exercise 0.32 implies that $\phi = 0$. □

Definition 2.4. Suppose $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$. Then the *total derivative*, or just *derivative*, of f at a, denoted df_a, is the unique linear map $\mathbb{R}^m \to \mathbb{R}^n$ satisfying

$$|f(a+v) - f(a) - df_a(v)| = o(|v|) \text{ as } v \to 0.$$

This definition is very difficult to compute examples from. Virtually the only concrete example that we can calculate without much effort directly from the definition is the following.

Exercise 2.5. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. Prove that ℓ is differentiable at every $a \in \mathbb{R}^m$ and $d\ell_a = \ell$.

Exercise 2.6. If $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$, show that f must be continuous at a.

2.C. Computing derivatives

2.C.1. Sum and scalar multiples rule

Exercise 2.7 (Sum rule)**.** Prove that, if $f, g : S \to \mathbb{R}^n$ are both differentiable at an interior point $a \in S$, then $f + g$ is also differentiable at a and

$$d(f+g)_a = df_a + dg_a.$$

Exercise 2.8 (Scalar multiples rule)**.** Prove that, if c is a constant and $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$, then cf is also differentiable at a and

$$d(cf)_a = c \cdot df_a.$$

2.C.2. Chain rule

The proof of the following is essentially the same as the second proof of the single variable chain rule given in section 1.B.3.B. Since this proof is basically a repeat, some details are omitted; you are asked to fill in the details in exercise 2.11 below.

Proposition 2.9 (Chain rule)**.** *Suppose that S and T are subsets of $\mathbb{R}^m$ and $\mathbb{R}^n$, respectively, that $f : S \to T$ is differentiable at an interior point a, that $f(a)$ is an interior point of T, and that $g : T \to \mathbb{R}^p$ is differentiable at $f(a)$. Then the composite $g \circ f : S \to \mathbb{R}^p$ is also differentiable at a, and*

$$d(g \circ f)_a = dg_{f(a)} \circ df_a.$$

Proof. Define the "error in approximation" functions

$$\begin{aligned} r(v) &= f(a+v) - f(a) - df_a(v) \\ s(w) &= g(f(a)+w) - g(f(a)) - dg_{f(a)}(w) \end{aligned}$$

and then observe that

$$g(f(a+v)) - g(f(a)) - dg_{f(a)}(df_a(v)) = dg_{f(a)}(r(v)) + s(df_a(v) + r(v)). \tag{2.10}$$

This means that

$$|g(f(a+v)) - g(f(a)) - dg_{f(a)}(df_a(v))| \leqslant |dg_{f(a)}(r(v))| + |s(df_a(h) + r(v))|.$$

Since $dg_{f(a)}$ is linear and $|r(v)| = o(|v|)$, exercise 0.30 guarantees that $|dg_{f(a)}(r(v))| = o(|h|)$ also. Thus it is sufficient to prove that $|s(df_a(v) + r(v))| = o(|v|)$. To do this, define $\eta(w) = |s(w)|/|w|$ and then notice that

$$\begin{aligned}\frac{|s(df_a(v) + r(v))|}{|v|} &= \eta(df_a(v) + r(v)) \cdot \frac{|df_a(v) + r(v)|}{|v|} \\ &\leqslant \eta(df_a(v) + r(v)) \left(\|df_a\| + \frac{|r(v)|}{|v|} \right).\end{aligned}$$

We now take the limit as $h \to 0$. Using the facts that $\eta \circ (df_a + r)$ is continuous at 0 and that $|r(v)| = o(|v|)$, we obtain the result. □

Exercise 2.11. (a) Prove equation (2.10).

(b) If you haven't done it already, do exercise 0.30.

(c) Prove that $\eta \circ (df_a + r)$ is continuous at 0.

2.C.3. Differentiability by components

Here we show that it's "enough" to consider the case $n = 1$.

Definition 2.12. For $j = 1, \ldots, n$, let $\pi_j : \mathbb{R}^n \to \mathbb{R}$ denote the jth *projection map*

$$\pi_j(x_1, \ldots, x_n) = x_j.$$

Then, for any function $f : S \to \mathbb{R}^n$, we define the jth *component function* $f_j : S \to \mathbb{R}$ to be the composite $\pi_j \circ f$.

Lemma 2.13. *A function $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$ if and only if the component function $f_j : S \to \mathbb{R}$ is differentiable at a for all $j = 1, \ldots n$. Moreover,*

$$df_a(v) = \begin{bmatrix} df_{1,a}(v) \\ \vdots \\ df_{n,a}(v) \end{bmatrix}. \tag{2.14}$$

Proof. If f is differentiable at a, then the composite $f_j = \pi_j \circ f$ is also differentiable at a by the chain rule 2.9. Moreover, since π_j is linear, we know that $d\pi_{j,f(a)} = \pi_j$ by exercise 2.5. Thus, by the chain rule,

$$df_{j,a}(v) = (d\pi_{j,f(a)} \circ df_a)(v) = \pi_j(df_a(v))$$

for all j, which is precisely equation (2.14). For the converse, we turn equation (2.14) on its head. Suppose f_j is differentiable at a for all j, and let $\ell : \mathbb{R}^m \to \mathbb{R}^n$ be the linear map

$$\ell(v) = \begin{bmatrix} df_{1,a}(v) \\ \vdots \\ df_{n,a}(v) \end{bmatrix}.$$

We now want to show that $|f(a+v) - f(a) - \ell(v)| = o(|v|)$ as $v \to 0$. Observe that the component functions of $v \mapsto f(a+v) - f(a) - \ell(v)$ are given by

$$v \mapsto \pi_j\left(f(a+v) - f(a) - \ell(v)\right) = f_j(a+v) - f_j(a) - df_{j,a}(v),$$

and we know that $|f_j(a+v) - f_j(a) - df_{j,a}(v)| = o(|v|)$. By exercise 2.15 below applied with the function $r(v) = f(a+v) - f(a) - \ell(v)$, we conclude that $|f(a+v) - f(a) - \ell(h)| = o(|v|)$. $\square$

Exercise 2.15. (a) For any vector $w \in \mathbb{R}^n$, show that

$$|w| \leqslant \sqrt{n} \max_j |\pi_j(w)|.$$

(b) Show that, if S is a neighborhood of 0 in $\mathbb{R}^m$ and $r : S \to \mathbb{R}^n$ is a function such that $|r_j(v)| = o(|v|)$ as $v \to 0$ for all $j = 1, \ldots, n$, then $|r(v)| = o(|v|)$ also. *Hint.* Use part (a).

2.C.4. Partial derivatives

Here we focus on the case $n = 1$. In other words, we consider functions whose input is m real numbers, and whose output is just a single number.

Definition 2.16. Suppose $f : S \to \mathbb{R}$ is a function and $a \in S$ is an interior point. The ith *partial derivative* of f at a, denoted $\partial_i f(a)$, is the limit

$$\lim_{h \to 0} \frac{f(a + he_i) - f(a)}{h}.$$

In other words, consider the function $\xi_i : \mathbb{R} \to \mathbb{R}^m$ given by $\xi_i(h) = a + he_i$. Then 0 is an interior point of $T = \xi_i^{-1}(S) \subseteq \mathbb{R}$, and $\partial_i f(a)$ is simply the derivative of the single variable function $f \circ \xi_i : T \to \mathbb{R}$ at 0.

Lemma 2.17. *If $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$, then the ith partial derivative $\partial_i f(a)$ exists for all $i = 1, \dots m$, and the standard matrix representation $[df_a]$ of the linear map $df_a : \mathbb{R}^m \to \mathbb{R}$ is a $1 \times m$ matrix whose ith entry is $\partial_i f(a)$.*

$$[df_a] = \begin{bmatrix} \partial_1 f(a) & \partial_2 f(a) & \cdots & \partial_m f(a) \end{bmatrix}$$

Proof. Let r be the "error in approximation" function

$$r(v) = f(a + v) - f(a) - df_a(v),$$

so that $|r(v)| = o(|v|)$ as $h \to 0$. Then

$$\begin{aligned} 0 &= \lim_{h \to 0} \frac{|r(he_i)|}{|he_i|} \\ &= \lim_{h \to 0} \frac{|f(a + he_i) - f(a) - df_a(he_i)|}{|h|} \\ &= \lim_{h \to 0} \left| \frac{f(a + he_i) - f(a)}{h} - df_a(e_i) \right| \end{aligned}$$

which means that $df_a(e_i) = \partial f_a(e_i)$. □

Remark 2.18. If $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$, then the vector

$$[df_a]^T = \begin{bmatrix} \partial_1 f(a) \\ \partial_2 f(a) \\ \vdots \\ \partial_m f(a) \end{bmatrix} \in \mathbb{R}^m$$

is often called the *gradient* of f at a, and is denoted $\nabla f(a)$.

Unfortunately, the converse to lemma 2.17 is *not* true, as shown by exercise 2.19. However, this behavior is fairly pathological; theorem 2.27 below will provide a converse under a fairly mild additional hypothesis.

Exercise 2.19. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined as follows.

$$f(x_1, x_2) = \begin{cases} \dfrac{x_1 x_2}{x_1^2 + x_2^2} & \text{if } (x_1, x_2) \neq 0 \\ 0 & \text{if } (x_1, x_2) = 0 \end{cases}$$

Show that $\partial_1 f(0)$ and $\partial_2 f(0)$ both exist, but that f is not even continuous at 0, so cannot be differentiable by exercise 2.6. *Hint.* Approach the origin along the line $x_1 = x_2$.

2.C.5. Jacobian matrix

Definition 2.20. Suppose $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$. The *Jacobian matrix* at a, denoted $J(a)$, is the $n \times m$ matrix whose (j, i)-entry is $\partial_i f_j(a)$. In other words,

$$J(a) = \begin{bmatrix} \partial_1 f_1(a) & \partial_2 f_1(a) & \cdots & \partial_m f_1(a) \\ \partial_1 f_2(a) & \partial_2 f_2(a) & \cdots & \partial_m f_2(a) \\ \vdots & \vdots & \ddots & \vdots \\ \partial_1 f_n(a) & \partial_2 f_n(a) & \cdots & \partial_m f_n(a) \end{bmatrix}$$

Note that lemma 2.17 tells us that all of these partial derivatives exist.

Theorem 2.21. *Suppose $f : S \to \mathbb{R}^m$ is differentiable at an interior point $a \in S$. Then*

$$[df_a] = J(a).$$

Exercise 2.22. Prove theorem 2.21. *Hint.* Use lemmas 2.13 and 2.17.

2.D. Levels of differentiability

Throughout, U denotes an open subset of $\mathbb{R}^m$.

Definition 2.23. If a function $f : U \to \mathbb{R}^n$ is differentiable at every $a \in U$, then f is *differentiable on* U. Then the *derivative* of f, denoted df, is the function $U \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ given by $a \mapsto df_a$. Given $i = 1, \ldots, m$ and $j = 1, \ldots n$, the i*th partial derivative* $\partial_i f_j$ is the function $U \to \mathbb{R}$ defined by $a \mapsto \partial_i f_j(a)$.

2.D.1. Continuous differentiability

This section is surprisingly challenging. You may need to read through it several times. My suggestion is that you skip the proofs the first time around, wrap your head around the big picture, and then fill in the details by working through the proofs.

Recall from definition 0.26 that the set $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ of all linear maps $\mathbb{R}^m \to \mathbb{R}^n$ is a metric space under the operator norm. We use this metric space structure to make the following definition.

Definition 2.24. If $f : U \to \mathbb{R}^n$ is a differentiable function and the derivative $df : U \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ is continuous, then f is said to be *continuously differentiable*.

Lemma 2.25. *A function* $f : U \to \mathbb{R}^n$ *is continuously differentiable if and only if the component functions* $f_j : U \to \mathbb{R}$ *are all continuously differentiable for* $j = 1, \ldots, n$.

Proof. We will use lemma 2.13, which tells us that f is differentiable if and only if f_j is differentiable for all $j = 1, \ldots, n$, and moreover that $df_{j,a} = \pi_j \circ df_a$ for all $a \in U$. In other words, the following diagram commutes.

$$\begin{tikzcd} U \arrow[r, "df"] \arrow[dr, bend right, "df_j"'] & \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \arrow[d, "\pi_j \circ -"] \\ & \mathcal{L}(\mathbb{R}^m, \mathbb{R}) \end{tikzcd}$$

You will verify in exercise 2.26(a) below that the map "postcompose with π_j" map $\pi_j \circ - : \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \to \mathcal{L}(\mathbb{R}^m, \mathbb{R})$ is continuous. Thus, if f is continuously differentiable, then df_j is the composite of the two continuous maps df and $\pi_j \circ -$, so it is itself continuous.

Conversely, suppose that df_j is continuous for all j. Let us show that df is continuous at every $a \in U$. Suppose $\epsilon > 0$. Since df_j is continuous, there exists $\delta_j > 0$ such that

$$\|df_{j,a} - df_{j,b}\| < \frac{\epsilon}{\sqrt{n}}$$

for all $b \in U$ such that $|a - b| < \delta_j$. Let $\delta := \min\{\delta_1, \ldots, \delta_n\}$. Then $\|df_a - df_b\| < \epsilon$ for all $b \in U$ such that $|a - b| < \delta$, as you will verify in exercise 2.26(b) below. □

Exercise 2.26. (a) Check that the "postcompose with π_j" map $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \to \mathcal{L}(\mathbb{R}^m, \mathbb{R})$ (ie, the one given by $\phi \mapsto \pi_j \circ \phi$) is continuous. Alternatively, prove the more general statement given in exercise 0.29 (which is not really much harder).

(b) With notation as in the latter part of the proof above, verify that $\|df_a - df_b\| < \epsilon$.

Theorem 2.27. *A function* $f : U \to \mathbb{R}^n$ *is continuously differentiable if and only if the partial derivatives* $\partial_i f_j : U \to \mathbb{R}$ *exist and are continuous for all* $i = 1, \ldots, m$ *and* $j = 1, \ldots, n$.

Proof. Thanks to lemma 2.25, we can assume without loss of generality that $n = 1$. If f is differentiable, then lemma 2.17 tells us that $\partial_i f = df(e_i)$. In other words, the following diagram commutes.

$$\begin{tikzcd} U \arrow[r, "df"] \arrow[dr, bend right, "\partial_i f"'] & \mathcal{L}(\mathbb{R}^m, \mathbb{R}) \arrow[d, "\text{“evaluate at } e_i\text{”}"] \\ & \mathbb{R} \end{tikzcd}$$

The "evaluate at e_i" map is continuous by exercise 0.27. If f is continuously differentiable, then df is continuous, so we conclude that the composite $\partial_i f$ is also continuous.

The converse is the hard part of this proof. Recall from exercise 2.19 above that $\partial_i f$ existing for all i is *not* sufficient to guarantee differentiability; we will need to use the fact that $\partial_i f$ exists *and is continuous* to conclude differentiability.

Fix $a \in U$. Let ℓ be the linear map $\mathbb{R}^m \to \mathbb{R}$ such that

$$[\ell] = \begin{bmatrix} \partial_1 f(a) & \cdots & \partial_m f(a) \end{bmatrix}.$$

In other words, ℓ is given by

$$\ell(h_1, \ldots, h_m) = h_1 \partial_1 f(a) + \cdots + h_m \partial_m f(a).$$

We will show that

$$|f(a+v) - f(a) - \ell(v)| = o(|v|) \text{ as } v \to 0,$$

which implies that f is differentiable at a and that $df_a = \ell$. In other words, we will show that, for any $\epsilon > 0$, there exists a $\delta > 0$ such that, for all $v \in \mathbb{R}^m$ with $|v| < \delta$, we have

$$|f(a+v) - f(a) - \ell(v)| \leqslant |v|\epsilon. \tag{2.28}$$

Let $\delta_0 > 0$ be small enough that the open ball $B(a, \delta_0)$ is contained in U. We will soon make it smaller to get inequality (2.28) to hold. Suppose $v = (h_1, \ldots, h_m) \in \mathbb{R}^m$ with $|v| < \delta_0$. Then consider the following vectors in $\mathbb{R}^m$.

$$\begin{aligned} v_0 &= 0 \\ v_1 &= (h_1, 0, 0 \ldots, 0, 0) \\ v_2 &= (h_1, h_2, 0, \ldots, 0, 0) \\ &\vdots \\ v_{m-1} &= (h_1, h_2, h_3, \ldots, h_{m-1}, 0) \\ v_m &= (h_1, h_2, h_3, \ldots, h_{m-1}, h_m) = v \end{aligned}$$

See figure 2.29. Clearly $|v_i| \leqslant |v|$ for all i.

Notice that

$$f(a+v) - f(v) = \sum_{i=1}^{m} \left(f(a+v_i) - f(a+v_{i-1}) \right).$$

Let us fix i temporarily. Notice that $v_i = v_{i-1} + h_i e_i$, so

$$f(a+v_i) - f(a+v_{i-1}) = f(a+v_{i-1}+h_i e_i) - f(a+v_{i-1}).$$

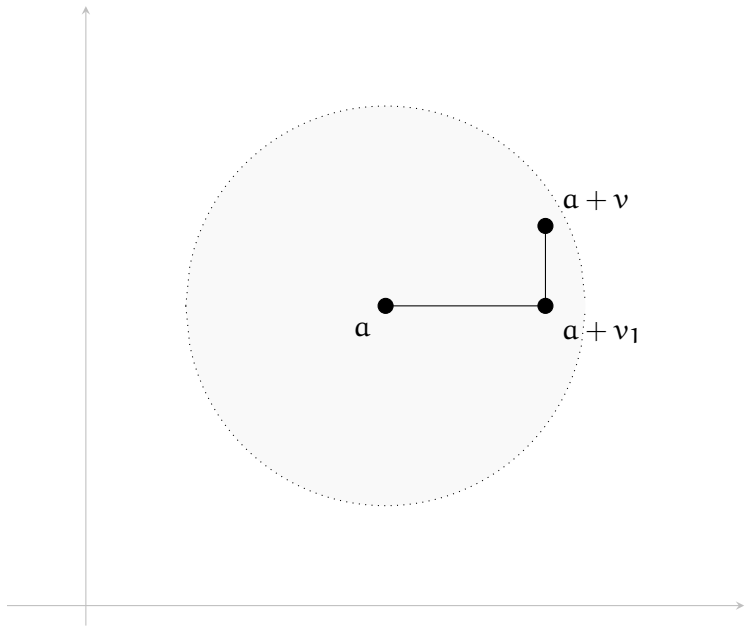

Figure 2.29.: This is a picture of an open ball of some radius δ_0 around a point $a \in \mathbb{R}^2$. The vector $v \in \mathbb{R}^2$ has $|v| < \delta$, so that $a + v$ is a point inside this ball. Defining v_1 as in the proof of theorem 2.27, the vector $a + v_1$ is also inside this ball. Then, defining s_1 and s_2 as in the proof, the point $a + s_1 e_1$ is somewhere along the horizontal line joining a and $a + v_1$, and the point $a + v_1 + s_2 e_2$ is somewhere on the vertical line joining $a + v_1$ and $a + v$.

We now apply the mean value theorem 1.35 to the single variable function

$$h \mapsto f(a + v_{i-1} + he_i)$$

on the closed interval between 0 and h_i. For any such h, we have $|v_{i-1} + he_i| < \delta$, so the points $a + v_{i-1} + he_i$ are all contained inside the ball $B(a, \delta_0) \subseteq U$. Thus this function is well-defined and continuous. It is also differentiable on the interior of this interval: its derivative at a value h is $\partial_i f(a + v_{i-1} + he_i)$, by definition of partial derivatives. The mean value theorem says that there exists s_i between 0 and h_i such that

$$\partial_i f(a + v_{i-1} + se_i) = \frac{f(a + v_{i-1} + h_i e_i) - f(a + v_{i-1})}{h_i} = \frac{f(a + v_i) - f(a + v_{i-1})}{h_i},$$

or, in other words,

$$f(a + v_i) - f(a + v_{i-1}) = h_i \partial_i f(a + v_{i-1} se_i).$$

See figure 2.29 again. Unfixing i, we find that

$$f(a + v) - f(a) = \sum_{i=1}^{m} (f(a + v_i) - f(a + v_{i-1})) = \sum_{i=1}^{m} h_i \partial_i f(a + v_{i-1} + s_i e_i).$$

Thus

$$|f(a + v) - f(a) - \ell(v)| \leqslant \sum_{i=1}^{m} |h_i| \cdot |\partial_i(a + v_{i-1} + s_i e_i) - \partial_i f(a)|.$$

Since $\partial_i f$ is continuous, there exists $\delta_i > 0$ such that $|\partial_i(a + w) - \partial_i(a)| < \epsilon/m$ for all $|w| < \delta_i$. Let $\delta = \min\{\delta_0, \delta_1, \ldots, \delta_m\}$. If $|v| < \delta$, then $|v_{i-1} + s_i e_i| < |v| < \delta$ for all i, so

$$|\partial_i(a + v_{i-1} + s_i e_i) - \partial_i f(a)| < \epsilon/m,$$

which means that

$$|f(a + v) - f(a) - \ell(v)| \leqslant \sum_{i=1}^{m} \frac{|h_i|\epsilon}{m} \leqslant |v|\epsilon.$$

This proves that f is differentiable at any point $a \in U$, which completes the hard part of this proof.

The final step is to show that $df : U \to \mathcal{L}(\mathbb{R}^m, \mathbb{R})$ is continuous. This is left for you to check; see exercise 2.30 below. □

Exercise 2.30. (a) Elaborate on the first sentence of the above proof. In other words, explain why proving theorem 2.27 for $n = 1$ is sufficient to prove it for all n.

(b) If you haven't already done it, do exercise 0.27.

(c) Complete the proof above by showing that $df : U \to \mathcal{L}(\mathbb{R}^m, \mathbb{R})$ is continuous.

2.D.2. Smooth functions

Definition 2.31. We make the following string of definitions.

- We say that a function $f : U \to \mathbb{R}^n$ is C^0 if it is continuous.
- Inductively, for any positive integer n, we say that a function $f : U \to \mathbb{R}^n$ is C^n if the partial derivatives $\partial_i f_j$ all exist and are C^{n-1} functions $U \to \mathbb{R}$.
- Finally, we say that a function $f : U \to \mathbb{R}^n$ is C^∞ or *smooth* if it is C^n for all n.

Theorem 2.27 tells us that $f : U \to \mathbb{R}^n$ is C^1 if and only if it is continuously differentiable.

2.E. Inverse function theorem

The following is a multivariable analog of exercise 1.28, but it is much harder to prove.

Theorem 2.32. *Suppose U is an open subset of $\mathbb{R}^m$ and $f : U \to \mathbb{R}^n$ is continuously differentiable. Suppose further that $a \in U$ is a point such that the derivative df_a is invertible. Then there exists an open neighborhood V of a in U such that $f(V)$ is open and the restriction $f|_V$ of f to V is an bijective function $V \to f(V)$. Moreover, if $g : f(V) \to V$ is the inverse of $f|_V$, then g is also continuously differentiable.*

3. Manifolds

At this point, we will need to use the language of topological spaces. If you have seen metric spaces but not topological spaces, you should look over section 0.D.1 at this point.

3.A. Definition of a manifold

Throughout, let X be a topological space.

Definition 3.1. A *chart* on X is a pair (U, x) consisting of an open subset $U \subseteq X$ and a function $x : U \to \mathbb{R}^n$ which is a homeomorphism onto an open subset of $\mathbb{R}^n$ for some n. We call the functions $x_i := \pi_i \circ x : U \to \mathbb{R}$ *coordinate functions* on U with respect to the chart (U, x), and for any $P \in U$, the list of numbers $x_1(P), \ldots, x_n(P)$ are the *coordinates* of P with respect to the chart (U, x).

Exercise 3.2. Let $X = \mathbb{R}^2$ and let U denote $\mathbb{R}^2$ minus the non-negative real axis. Let $\text{polar} : U \to \mathbb{R}^2$ be the function which sends a point P to its polar representation $(r(P), \theta(P))$, where $r(P) = |P|$ and $\theta(P)$ is the angle in radians strictly between 0 and 2π formed between P and the non-negative real axis. Show that this function is a homeomorphism onto its image.

Definition 3.3. Suppose (U, x) and (V, y) are charts on X. The *transition map* from (U, x) to (V, y) is the function $y \circ x^{-1}$.

$$x(U \cap V) \xrightarrow{x^{-1}} U \cap V \xrightarrow{y} y(U \cap V)$$

Definition 3.4. The two charts (U, x) and (V, y) on X are *smoothly compatible*, or just *compatible*, if both of the transition maps between (U, x) and (U, y) are smooth.

Exercise 3.5. Let $\text{id} : \mathbb{R}^2 \to \mathbb{R}^2$ denote the identity map and $\text{polar} : U \to \mathbb{R}^2$ the polar representation map from exercise 3.2. Show that $(\mathbb{R}^2, \text{id})$ and (U, polar) are compatible charts.

Exercise 3.6. Show that the relation "is compatible with" is *not* an equivalence relation on the set of all charts on X. *Hint*. The function $\mathbb{R} \to \mathbb{R}$ given by $x \mapsto x^3$ is a smooth homeomorphism, but its inverse is not differentiable.

Exercise 3.7. Suppose (U, x) and (V, y) are compatible charts, where $x : U \to \mathbb{R}^m$ and $y : V \to \mathbb{R}^n$, and that $U \cap V$ is nonempty. Show that $m = n$. *Hint*. Pick a point $P \in U \cap V$, and consider the derivatives of $x \circ y^{-1}$ at $y(P)$ and of $y \circ x^{-1}$ at $x(P)$.

Definition 3.8. An *atlas* on X is a collection $\mathcal{A}$ of charts such that covers X. In other words,

$$\bigcup_{(U,x)\in\mathcal{A}} U = X.$$

An atlas is *maximal* if there is no strictly larger atlas containing it.

Exercise 3.9. Prove that every atlas $\mathcal{A}$ is contained in a unique maximal atlas. *Hint*. Let $\mathcal{A}'$ be the set of all charts that are compatible with all of the charts in $\mathcal{A}$. Prove that $\mathcal{A}'$ is an atlas (caution: keep exercise 3.6 in mind). Then prove that it is maximal, and it is the only maximal atlas containing $\mathcal{A}$.

This leads us to the following definition.

Definition 3.10. A *smooth manifold*, or just a *manifold*, is a pair $(X, \mathcal{A}_X)$ where X is a topological space and $\mathcal{A}_X$ is a maximal atlas. Often, write simply "X" in place of the pair $(X, \mathcal{A}_X)$, and sometimes $\mathcal{A}_X$ is called the "manifold structure on X."

Of course, exercise 3.9 tells us that any not-necessarily-maximal atlas on a metric space X defines a manifold structure on X in a canonical way. When constructing examples, it is usually useful to consider non-maximal atlases. But, when dealing with abstract manifolds, we will often find it convenient to assume that the atlas that the manifold comes equipped with is already maximal, which is the reason that the definition is made the way it is.

3.B. Examples

3.B.1. First examples

Example 3.11. The single chart $(\mathbb{R}^n, \mathrm{id})$, where $\mathrm{id} : \mathbb{R}^n \to \mathbb{R}^n$ is the identity map, defines an atlas on $\mathbb{R}^n$. We can extend this to a maximal atlas by exercise 3.9, called the *euclidean atlas* on $\mathbb{R}^n$. Unless explicitly specified otherwise, we always regard $\mathbb{R}^n$ as a manifold by equipping it with the euclidean atlas. By exercise 3.5, the polar representation chart (U, polar) is an element of the euclidean atlas.

Exercise 3.12. The single chart $(\mathbb{R}, f)$ where $f : \mathbb{R} \to \mathbb{R}$ is the function $f(x) = x^3$ is an atlas on $\mathbb{R}$. Show that the corresponding maximal atlas is *not* the euclidean atlas.

Exercise 3.13 (The circle S^1)**.** Let S^1 denote the set of points in $\mathbb{R}^2$ along the unit circle. In other words,

$$S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

To regard S^1 as a topological space, we do what you might expect: we first restrict the euclidean metric on $\mathbb{R}^2$ to S^1, and then we forget this restricted metric and just remember the resulting topology as in example 0.34.

Given a point $P \in S^1 \setminus \{(0, 1)\}$, the *projection of* P *from* $(0, 1)$ *onto* $\mathbb{R} \times \{-1\}$, which we will denote $\pi_1(P)$, is defined as the point of intersection between the line $y = -1$ and the line passing through P and $(0, 1)$. See figure 3.15. Similarly, for $P \in S^1 \setminus \{(0, -1)\}$, we can define the projection of P from $(0, -1)$ onto $\mathbb{R} \times \{1\}$, denoted $\pi_2(P)$. Show that π_1 and π_2 together define an atlas on S^1.

Exercise 3.14 (The n-sphere S^n)**.** Let S^n denote the set of points in $\mathbb{R}^{n+1}$ of norm 1. Generalize exercise 3.13 and show that "projection from the north pole" and "projection from the south pole" define an atlas on S^n.

Exercise 3.16 (Open submanifolds)**.** Let X be a manifold and $U \subseteq X$ an open subset. Show that

$$\mathcal{A}_U = \{(V, x) \in \mathcal{A}_X : V \subseteq U\}$$

is a maximal atlas on U. We call U, equipped with this maximal atlas, an *open submanifold* of X.

Exercise 3.17 (Product manifold)**.** Let X and Y be manifolds. Let $\mathcal{A}$ be the set of charts of the form $(U \times V, x \times y)$, where $(U, x : U \to \mathbb{R}^m) \in \mathcal{A}_X$, $(V, y : V \to \mathbb{R}^n) \in \mathcal{A}_Y$, and $x \times y$ denotes the natural map $U \times V \to \mathbb{R}^m \times \mathbb{R}^n = \mathbb{R}^{m+n}$. Show that $\mathcal{A}$ is an atlas on $X \times Y$. We call $X \times Y$, equipped with the corresponding maximal atlas, the *product manifold*.

3.B.2. Möbius strip

Let B denote the box $[0, 1] \times (0, 1)$ inside $\mathbb{R}^2$, and define an equivalence relation $\sim$ on B by declaring $(0, b) \sim (1, 1 - b)$ for all $b \in (0, 1)$. We will denote the equivalence class of a point $(a, b) \in B$ as $[a, b]$.

Exercise 3.18. Check that $\sim$ is an equivalence relation.

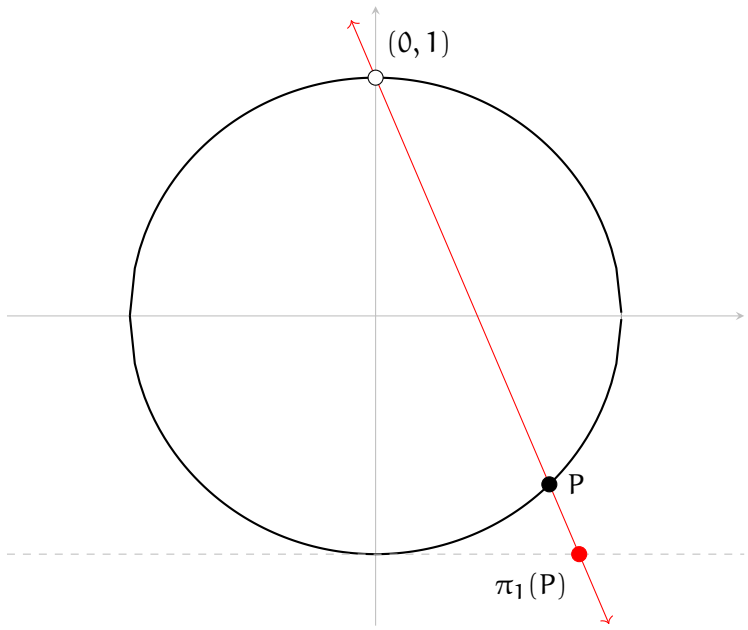

Figure 3.15.: Given a point P on $S^1 \setminus \{(0,1)\}$, we draw a line connecting P to $(0,1)$. The point where that line intersects the line $\mathbb{R} \times \{-1\}$ is the *projection of* P *from* $(0,1)$ *onto* $\mathbb{R} \times \{-1\}$, denoted $\pi_1(P)$.

The *Möbius strip* is the set of equivalence classes $M := B/\sim$. In other words, we take the box $[0,1] \times (0,1)$ and "glue" its left and right edges, with a twist. If you've ever made a Möbius strip out of paper, you should be able to see that we've formalized exactly that same process. We regard M as a topological space using the quotient topology, as in example 0.35.

To give X the structure of a manifold, we will now produce an atlas on X. It will consist of two charts. The first chart is easy to describe. Let

$$U = \{[a,b] \in M : a \neq 0, 1\}$$

and let $x : U \to \mathbb{R}^2$ be the function $x([a,b]) = (a,b)$.

Exercise 3.19. Show that $x : U \to \mathbb{R}^2$ is well-defined and that it is a homeomorphism onto its image.

Intuitively, what we've done here is cut along the same edge where we glued the Mobius strip together. What we're left with is U, which is just a flat rectangle and therefore embeds into $\mathbb{R}^2$.

The second chart is a bit harder to describe mathematically, but the intuitive idea is the same. Instead of cutting the Möbius strip along the line where we glued, we'll cut along the line where $a = 1/2$. This should still leave us with a flat rectangle, which we can embed into $\mathbb{R}^2$. Formally, let

$$V = \{[a,b] \in M : a \neq 1/2\}$$

and let $y : V \to \mathbb{R}^2$ be the function defined by

$$y([a,b]) = \begin{cases} (a,b) & \text{if } a < 1/2 \\ (a-1, 1-b) & \text{if } a > 1/2 \end{cases}$$

Exercise 3.20. Show that $y : V \to \mathbb{R}^2$ is well-defined and that it is a homeomorphism onto its image.

Exercise 3.21. Show that (U,x) and (V,y) are compatible.

Thus $\mathcal{A} = \{(U,x),(V,y)\}$ is an atlas on the Möbius strip M. Letting $\mathcal{A}_M$ denote the corresponding maximal atlas, we obtain the structure of a manifold on M.

3.B.3. More examples of manifolds

Exercise 3.22 (Torus)**.** Let B denote the box $[0,1] \times [0,1]$ inside $\mathbb{R}^2$ and define the equivalence relation $\sim$ by $(a,0) \sim (a,1)$ and $(0,b) \sim (1,b)$ for all $a, b \in [0,1]$. The *torus* T is defined to be the quotient $B/\sim$. Describe an atlas on T.

Exercise 3.23 (Projective line)**.** Let S^1 denote the circle as in exercise 3.13. Define an equivalence relation $\sim$ on S^1 which identifies antipodal points; in other words, every point on S^1 is declared to be equivalent to the point that is diametrically opposite it. The *projective line* $\mathbb{P}^1$ is defined to be the quotient $S^1/\sim$. Describe an atlas on $\mathbb{P}^1$.

Exercise 3.24 (Projective plane)**.** Let S^2 denote the 2-sphere as in exercise 3.14. Define an equivalence relation $\sim$ on S^2 which identifies antipodal points; in other words, every point on S^2 is declared to be equivalent to the point that is diametrically opposite it. The *projective plane* $\mathbb{P}^2$ is defined to be the quotient $S^2/\sim$. Describe an atlas on $\mathbb{P}^2$.

3.C. Smooth functions

3.C.1. Smooth real-valued functions

Definition 3.25. Let X be a manifold. A function $f : X \to \mathbb{R}$ is said to be *smooth* if, for any chart $(U, x) \in \mathcal{A}_X$, the function $f \circ x^{-1}$ is smooth.

$$x(U) \xrightarrow{x^{-1}} U \xrightarrow{f} \mathbb{R}$$

There are two disconcerting things about this definition. First, there is the issue that checking smoothness of a particular function seems next to impossible: the maximal atlas $\mathcal{A}_X$ will have *many, many* charts, and the definition makes it seem like we need to check smoothness on each of these charts.

Thankfully, we don't need to check on all possible charts. As soon as we check smoothness on a collection of charts that cover X, we're done.

Lemma 3.26. *Let X be a manifold and $f : X \to \mathbb{R}$ a function. Suppose that there exists an atlas $\mathcal{A} \subseteq \mathcal{A}_X$ such that $f \circ x^{-1}$ is smooth for any $(U, x) \in \mathcal{A}$. Then f is smooth.*

Proof. Suppose $(V, y) \in \mathcal{A}_X$. For any $(U, x) \in \mathcal{A}$, we have maps as follows.

$$\begin{tikzcd}
y(U \cap V) \arrow[r, "y^{-1}"] \arrow[d, "x \circ y^{-1}"'] & U \cap V \arrow[r, "f"] & \mathbb{R} \\
x(U \cap V) \arrow[ur, bend right, "x^{-1}"'] & &
\end{tikzcd}$$

Since (V, y) and (U, x) are both elements of the maximal atlas $\mathcal{A}_X$, we know that the transition function $x \circ y^{-1}$ is smooth. Since $(U, x) \in \mathcal{A}$, we know by assumption that $f \circ x^{-1}$ is smooth. Thus

$$f \circ y^{-1} = (f \circ x^{-1}) \circ (x \circ y^{-1})$$

is a smooth function $y(U \cap V) \to \mathbb{R}$.

Now note that, since $\mathcal{A}$ is an atlas, its charts cover all of X, so

$$\bigcup_{(U,x)\in\mathcal{A}} y(U \cap V) = y(V).$$

Since $f \circ y^{-1}$ is smooth when restricted to each the open subset $y(U \cap V)$ for all $(U, x) \in \mathcal{A}$, we conclude that $f \circ y^{-1}$ must be smooth on all of $y(V)$. □

The second disconcerting thing is the following. Suppose $U \subseteq \mathbb{R}^m$ is an open subset and $f : U \to \mathbb{R}$ is a function. On the one hand, we defined what it means for f to be smooth back in definition 2.31. On the other hand, we can regard U as an open submanifold of $\mathbb{R}^n$ (cf. exercise 3.16) and then definition 3.25 gives us another definition of what it means to be smooth. A priori, it could be that these definitions conflict. Thankfully, they do not.

Exercise 3.27. Suppose $U \subseteq \mathbb{R}^m$ is an open subset and $f : U \to \mathbb{R}$ is a function. Show that f is smooth in the sense of definition 2.31 if and only if it is smooth in the sense of definition 3.25. *Hint*. You might find lemma 3.26 useful.

3.C.2. Partial derivatives of real-valued functions

Definition 3.28. Let X be a manifold and $(U, x) \in \mathcal{A}_X$ a chart. Recall that we then have coordinate functions $x_1, \ldots, x_n : U \to \mathbb{R}$, where $x_i = \pi_i \circ x$. Given a smooth function $f : X \to \mathbb{R}$ and a point $P \in U$, we define the *partial derivative of* f *at* P *with respect to* x_i by

$$\left.\frac{df}{dx_i}\right|_P := \partial_i(f \circ x^{-1})(x(P)).$$

In other words, $df/dx_i = \partial_i(f \circ x^{-1}) \circ x$ as functions $U \to \mathbb{R}$.

3.C.3. Smooth functions between manifolds

Definition 3.29. Suppose X and Y are manifolds and $f : X \to Y$ is a function. Then f is *smooth* if, for every chart $(V, y) \in \mathcal{A}_Y$ and every chart $(U, x) \in \mathcal{A}_X$ such that $f(U) \subseteq V$, the

function $y \circ f \circ x^{-1} : x(U) \to y(V)$ is smooth.

$$\begin{tikzcd}
X \arrow[r, "f"] & Y \\
U \arrow[u, hook] \arrow[r, "f"] \arrow[d, "x"'] & V \arrow[u, hook] \arrow[d, "y"] \\
x(U) \arrow[r, "y \circ f \circ x^{-1}"] \arrow[d, hook] & y(V) \arrow[d, hook] \\
\mathbb{R}^m & \mathbb{R}^n
\end{tikzcd}$$

Once again, we don't need to check all possible charts. The precise statement is a bit complicated though; there's a long string of quantifiers.

Exercise 3.30. Suppose X and Y are manifolds and $f : X \to Y$ is a function. Suppose further that there exists an atlas $\mathcal{A} \subseteq \mathcal{A}_Y$ such that, for every $(V, y) \in \mathcal{A}$, there exists an atlas $\mathcal{A}' \subseteq \mathcal{A}_{f^{-1}(V)}$, such that for every $(U, x) \in \mathcal{A}'$, the function $y \circ f \circ x^{-1} : x(U) \to y(V)$ is smooth. Show that f is smooth.

We should also note that we haven't redefined the word "smooth" in any conflicting ways. More precisely, we have the following two exercises.

Exercise 3.31. If $U \subseteq \mathbb{R}^m$ is an open subset and $f : U \to \mathbb{R}^n$ is a function, show that f is smooth in the sense of definition 2.31 if and only if it is smooth in the sense of definition 3.29.

Exercise 3.32. Show that a function $f : X \to \mathbb{R}$ is smooth in the sense of definition 3.25 if and only if it is smooth in the sense of definition 3.29.

3.C.4. Diffeomorphisms

Definition 3.33. Suppose X and Y are manifolds. A function $f : X \to Y$ is a *diffeomorphism* if it is smooth, bijective, and the inverse function $f^{-1} : Y \to X$ is also smooth. If there exists a diffeomorphism $f : X \to Y$, then X and Y are *diffeomorphic*.

Exercise 3.34. Regard $\mathbb{R}$ as a manifold in two different ways: once with the euclidean atlas, and next with the maximal atlas containing $x \mapsto x^3$. Show that these two manifolds are diffeomorphic.

Exercise 3.35. If X is a manifold and $(U, x) \in \mathcal{A}_X$, prove that $x : U \to x(U)$ is a diffeomorphism.

4. Tangent Spaces

4.A. Tangent space

4.A.1. Curves based at a point

Definition 4.1. Let X be a manifold and $P \in X$ a point. A *curve in* X *based at* P is a smooth function $\gamma : (-1, 1) \to X$ such that $\gamma(0) = P$. If $(U, x) \in \mathcal{A}_X$ and $P \in U$, then $x^{-1} \circ \gamma$ is a function

Definition 4.2. Let X be a manifold and $P \in X$ a point. Two curves γ_1, γ_2 in X based at P are *equivalent* if, for all $(U, x) \in \mathcal{A}_X$ such that $P \in U$, we have

$$d(x^{-1} \circ \gamma_1)_0 = d(x^{-1} \circ \gamma_2)_0.$$

As usual, the first point of order is to check that we don't need to check all possible charts.

Exercise 4.3. Show that, if γ_1, γ_2 are two curves in X based at P and there exists a chart $(U, x) \in \mathcal{A}_X$ such that $P \in U$ and

$$d(x^{-1} \circ \gamma_1)_0 = d(x^{-1} \circ \gamma_2)_0,$$

then γ_1 is equivalent to γ_2.

Definition 4.4. For a manifold X and a point $P \in X$, we define the *tangent space of* X *at* P, denoted $T_P X$, to be the set of equivalence classes of curves in X based at P. Elements of $T_P X$ are called *tangent vectors* at P.

4.A.2. Vector space structure on the tangent space

4.A.3. Derivations

4.B. Pushing tangent vectors forward

Let $f : X \to Y$ be a smooth map of manifolds and $P \in X$ a point. Given a curve γ in X based at P, observe that $f \circ \gamma$ is a curve in Y based at $f(P)$.

Lemma 4.5. *If* $f : X \to Y$ *is a smooth map of manifolds,* $P \in X$*, and* γ_1 *and* γ_2 *are equivalent curves in* X *based at* P*, then* $f \circ \gamma_1$ *and* $f \circ \gamma_2$ *are equivalent curves in* Y *based at* $f(P)$*.*

This allows us to make the following definition, which in some sense is the "ultimate" generalization of derivatives.

Definition 4.6. Let $f : X \to Y$ be a smooth map of manifolds. Given a tangent vector $v \in T_P X$ defined by a curve γ, we define its *pushforward* $f_* v \in T_{f(P)} Y$ to be the tangent vector defined by $f \circ \gamma$. This defines a *pushforward map* $f_* : T_P X \to T_{f(P)} Y$.

Proposition 4.7. *Let* $f : X \to Y$ *be a smooth map of manifolds and* $P \in X$ *a point. Then* $f_* : T_P X \to T_{f(P)} Y$ *is a linear map.*

Bibliography